
Project Title	Guazhou Beidaqiao No.1 Wind Farm Project in Gansu Province, China
ERM CVS Project Reference	1940.V1
Client Name	Macquarie Bank Limited
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Version	Date
Version 01	28 January 2011 (Draft Validation Report)
Version 02	28 September 2011 (Final Validation Report)

Project Title: Guazhou Beidaqiao No.1 Wind Farm Project in Gansu Province, China			
Project Location		Guazhou County, Jiuquan City, Gansu Province	
Country:		P. R. China	
Project Parties		P. R. China, United Kingdom	
Project Participants		Macquarie Bank Limited. Hydrochina Guazhou Wind Power Co.,Ltd.	
Methodology used		ACM0002 "Consolidated baseline methodology for grid-connected electricity generation from renewable sources"	
Methodology version number		12.1.0	
Estimated Annual Average Emission Reductions		428,829 tCO ₂ e	
Crediting Period Dates		01 December 2011 to 30 November 2018	
GSP PDD Version	Date: 25 November 2010	Final PDD Version	Date: 08 June 2011
	Version Number: 1.0		Version Number: 3.0
	Start of GSP: 30 November 2010		

Summary:

ERM CVS was commissioned to validate the Guazhou Beidaqiao No.1 Wind Farm Project in Gansu Province, China on the basis of UNFCCC criteria for the Clean Development Mechanism (CDM), as well as criteria given to provide for consistent project operations, monitoring and reporting. UNFCCC criteria refer to Article 12 of the Kyoto Protocol, the CDM modalities and procedures and the subsequent decisions by the CDM Executive Board, including the Validation and Verification Manual.

The validation consisted of the following three phases: i) a desk review of the project design documents, ii) site assessment and follow-up interviews with project stakeholders and iii) the resolution of outstanding issues and the issuance of the final validation report and opinion.

Based on the work performed, it is ERM CVS's opinion that the project as described in the Project Design Document Version 3.0, dated 08 June 2011, meets all necessary criteria and requirements of the CDM, correctly applies the methodology ACM0002 version 12.1.0 and is expected to result in real, measurable and long term emission reductions. The DNA of the host Party has confirmed that the project assists in meeting sustainable development criteria.

ERM CVS therefore requests that the CDM Executive Board registers the project as a CDM project activity.

Client:	Macquarie Bank Limited		
Client Representative:	John Marlow		
Report approved by:		Signed	
Name Melanie Eddis			
Date: 28 September 2011			

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Abbreviations

BM	Build Margin
CAR	Corrective Action Request
CDM	Clean Development Mechanism
CDM EB	CDM Executive Board
CER	Certified Emission Reduction
CL	Clarification Request
CO ₂	Carbon dioxide
CO ₂ e	Carbon dioxide equivalent
COP	Conference of Parties
DNA	Designated National Authority
DOE	Designated Operational Entity
EIA	Environmental Impact Assessment
FSR	Feasibility Study Report
GHG	Greenhouse Gas
GSP	Global Stakeholder Process
FAR	Forward Action Request
IPCC	Intergovernmental Panel on Climate Change
IRR	Internal Rate of Return
LoA	Letter of Approval
MOP	Meeting of Parties
MP	Monitoring Plan
NDRC	National Development and Reform Committee
NWCPG	Northwest China Power Grid
ODA	Official Development Assistance
OM	Operating Margin
PDD	Project Design Document
PP	Project Participant
QC/QA	Quality control/Quality assurance
UNFCCC	United Nations Framework Convention on Climate Change
VVM	CDM Validation and Verification Manual

1. Introduction

1.1. Validation Objective

The purpose of a validation is to provide a thorough independent third party assessment of proposed CDM project activities to ensure that the proposed CDM project activity meets all the identified and applicable criteria for registration of projects under the Clean Development Mechanism. In particular, the project's baseline, additionality demonstration, applicability to an approved CDM methodology, monitoring plan (MP), and the project's compliance with relevant UNFCCC and host country criteria are validated in order to confirm that the project design as documented is sound and reasonable and meets the stated requirements and identified criteria. Validation is seen as necessary to provide assurance to stakeholders of the quality of the project and its intended generation of Certified Emission Reductions (CERs). UNFCCC criteria refer to the Kyoto Protocol criteria and the CDM rules and modalities and related decisions by the COP/MOP and the CDM Executive Board. The validation will result in a conclusion as to whether the project should be submitted to registration. The final decision on whether to register the project rests with Executive Board and the Parties involved.

1.2. Scope

The validation scope is defined as an independent and objective review of the Project Design Document (PDD) and associated documentation. The PDD and associated documentation is reviewed against the criteria and requirements stated in the CDM Validation and Verification Manual (VVM) (EB 55) and Article 12 of the Kyoto Protocol, the CDM modalities and procedures as agreed in the Marrakech Accords, as well as relevant decisions by the CDM Executive Board. The validation scope also included an assessment of completeness and accuracy of documentation, evaluation of evidences, information and assumptions made in the PDD and supporting documentation.

1.3. CDM Project Description

The proposed project is a new wind farm with an installed capacity of 201 MW (134*1.5MW). The wind farm will supply 461,464 MWh of electricity to the Northwest China Power Grid (referred to as 'NWCPG' or 'the grid') per year, which is dominated by fossil fuel fired power plants /4/.

The baseline scenario, which is the status-quo, is to provide equivalent electricity from the Grid whose emission factor is calculated according to the 'Tool to calculate the Emission Factor for an electricity system' version 02 /9/. Further details of the grid are validated in section 3.6.

The project description has been validated against the detailed Feasibility Study Report (FSR) of the project activity, developed by an independent third party accredited by the Chinese government - the 'Northwest Hydro Consulting Engineers' in January 2009 /4/. The FSR was approved by the National Development and Reform Commission (NDRC) on 21 April 2009 /4/. The project description was also cross-checked against the physical layout of the project observed during the site visit.

The accuracy and completeness of the project description is confirmed by ERM CVS.

1.4. Validation Personnel

Validation Team	Role	Coverage of sectoral scope	Coverage of technical area	Financial Expertise	Host country experience	Participated in site visit?
Wang Ning	Lead Validator	√	√		√	Yes
Li Wei	Lead Validator (supervisor)	√	√		√	No
Frank Chen	Validator	Partial	Partial		√	Yes
Fernanda Michalischen	Financial expert			√		No

Technical Review	Role	Coverage of sectoral scope	Coverage of technical area
Ying Li	Technical Reviewer	√	√
Jonathan Avis	Technical Reviewer (Supervisor)	√	√

Ning Wang is a lead GHG assessor based in Beijing. Mr. Wang Ning holds a Masters degree in Energy and Environmental Technology from the University of Høgskolen i Telemark, Norway, which included renewable energy. Mr. Wang Ning has more than two years of experience working as a GHG auditor and has worked on more than 40 CDM validation and verification projects in China, covering a wide range of technologies including hydro, wind, biomass, and waste energy recovery. Mr. Wang Ning's experience in the renewable energy sector includes participation in more than 25 audits in scope 1 (energy industries). Wang Ning has completed the ERM CVS CDM Validation and Verification training course.

Wei Li is a Lead Auditor based in Beijing with over 8 years experience in environmental management system assessment, more than two years experiences in CDM and VCS validation and verification projects, he has lead more than 70 CDM and VCS validation and verification projects in wind power, hydropower, waste heat recovery, coal mine methane, N₂O, biodiesel, natural

gas generation, solid waste incineration, etc. He has been responsible for fifteen CDM and VCS renewable validation projects in hydro and wind electricity generation plants, eight of them are wind projects..

Frank Chen is an auditor based in Guangzhou City, P.R. China. Mr Chen holds a bachelor's degree in Environmental Engineering from Guangdong University of Technology and he has completed the ERM CVS training course for CDM Validation & Verification. Mr Chen has over 5 years of experience in the areas of EHS due diligence, EHS compliance audits, environmental engineering and GHG auditing in China and South East Asia, including audits in the chemicals, oil and gas and energy sectors. Mr. Chen has conducted two CDM validation projects for wind farm projects in North China and three CDM validation projects for hydropower projects in Central China.

Fernanda Michalischen is a financial expert for ERM CVS and holds a Master of Science in business administration from the University of São Paulo, with a specialisation in Finance and Corporate Governance. Ms Michalischen has completed the ERM CVS training course on CDM validation and verification. She has worked at ERM since 2008, controlling the financial performance of projects and developing investment analyses of various projects and financial analyses for CDM projects. She is also a professor in finance at FIPECAFI/University of São Paulo. Her previous experience includes nearly 2 years as a research assistant at the University of São Paulo developing Pension Fund studies. She has experience assisting Strategic Committees (Financial, Tributary and Partnership issues), and developing financial viability studies of commercial acquisitions.

Ying Li is an environmental engineer with extensive practical experience in the carbon market. She has extensive experience in the development and implementation of CDM projects, developed over 4 years whilst working in the area of project-based mechanisms. Her background includes both project validation and verification in a number of sectoral scopes, including renewable energy, landfill, coal mine methane, waste heat recovery and natural gas generation. She has developed and managed the validation/verification of number of CDM project activities, including experiences in PDD development, the monitoring system building, data QA/QC, emission reduction calculation review and documentation review. Throughout this work she has obtained a good understanding of technical aspects in these sectoral scopes as well as their specific CDM aspects.

Jonathan Avis is CDM Business Manager for ERM CVS, and a GHG Auditor and Technical Reviewer with over 5 years experience in the carbon market. Mr. Avis holds an MSc in Environmental Management from the University of Oxford. His previous work experience includes screening and due diligence of carbon projects, Project Design Document (PDD) development, quality assurance and technical review of CDM project documentation, the development of carbon monitoring plans, and management of carbon projects through the validation, registration and verification stages. He has worked on the development and quality control of carbon projects in numerous sectors including hydroelectricity, wind energy, landfill gas, waste gas and heat, coal mine methane, biomass-to-energy and composting. Since joining ERM CVS Jonathan has worked as a Technical Reviewer and GHG auditor on numerous CDM validations.

2. Methodology

The validation was carried out in accordance with the VVM, version 01.2 (EB 55). The validation process employed standard auditing techniques and undertook necessary cross-checks and follow-up actions to ascertain the correctness of the information. The validation team included staff with experience in the relevant sectoral scopes and technical areas within the sectoral scope, and included local host country expertise, sectoral knowledge, and financial expertise. The validation report and associated documents have undergone a thorough technical review by ERM CVS before being submitted to the CDM Executive Board for registration. The validation consisted of the following key phases:

- I. Upload of the PDD for Global Stakeholder Process (GSP), receipt of any comments from stakeholders (GSP started on 30 November 2010)
- II. Desk review of documentation including PDD, methodology and key supporting documents and references
- III. A visit to the project site, including interviews with personnel responsible for developing the project (the site visit took place on 22 December 2010)
- IV. Development of a draft validation report, identifying non-compliances including Corrective Action Requests (CARs) and Clarification Requests (CLs), taking into account findings of the GSP, desk review and site visit / interviews
- V. Resolution of outstanding issues (CARs and CLs) and development of a final validation report and validation opinion

2.1. Global Stakeholder Process consultation

The original PDD was uploaded for global stakeholder comments. Relevant information can be found at <http://cdm.unfccc.int/Projects/Validation/DB/745Q6QJ71LT3UFAZRN5IEZUSZ3M66C/view.html>

No comments have been received during the period.

2.2. PDD and Additional Documentation Review

A detailed desk review of the PDD, methodology and all other associated documentation and references took place in advance of the site visit, and additional documents that were not available for the desk review were requested for review during the site visit.

The project information was also cross-checked through other sources in order to evaluate the authenticity and credibility of information and ascertain the correctness of the information against the requirements and established criteria. The review of the PDD at the desk review stage raised a number of issues which were further reviewed and validated through supporting documentation and cross-checking from other sources and interviewing relevant personnel involved in the project activity during the site visit

A list of all documents reviewed or referred to in the course of this validation is included in Appendix A. Document references throughout the report are given in a letter-based abbreviation format to reference to this document list, for example /4/ refers to the Feasibility Study Report document listed in the document list.

2.3. Site visit and Interviews

A site visit took place as described above. ERM CVS staff attending the site visit included Wang Ning (lead validator) and Frank Chen (validator). The site visit included a tour of the physical project site. The site visit also included a visit to the office of the project owner, Hydrochina Guazhou Wind Power Co., Ltd.

Representatives from the project owner, the CDM consultant and local stakeholder were interviewed, and document review took place during site visit. A list of interviewees can be found in appendix A.

The site visit enabled the validation team to undertake a detailed review of the project documentation, verify the supporting documentation, inspect the project site and verify its compliance with the PDD, cross-check the validity of the project information from other sources of information, and hold interviews with local stakeholders, relevant officials and other persons involved in the project activity.

2.4. Reporting

A checklist of the key requirements for validation is included as Appendix B 'CDM Validation Protocol Checklist'. It serves the following purposes:

- It organises, details and clarifies the requirements a CDM project activity is expected to meet.
- It ensures a transparent validation process where the validator will document how a particular requirement has been validated and the result of the validation.
- It must also list project components/issues not covered in the validation engagement

The CDM Validation Protocol Checklist describes the following:

Checklist Question	Reference	Comment	Draft Conclusion	Final Conclusion
The requirements that the project should meet	The documents used to check the answer to the checklist question	This section is used to elaborate and discuss the conformance to the checklist question, and to explain the conclusion reached. It includes the means of validation, which explains how conformance with the checklist is justified. For example document review (DR) or interview (I). N/A means not applicable	This is either acceptable based on evidence provided (OK), or a <i>Corrective Action Request</i> (CAR) is required due to non-compliance with the checklist question. A request for <i>Clarification</i> (CL) is used when the validation team has identified a need for further clarification. A 'Minor Issue' may be recorded for typographical errors or similar minor errors that do not have an impact on the compliance of the project to the CDM rules but nevertheless should be corrected to improve clarity. A <i>Forward Action Request</i> (FAR) could be raised for issues to be addressed during first verification that do not form part of the registration requirements	Indicates whether the CAR or CL has been closed out (OK).

Remediation Form

Clarification Requests (CL), Corrective Action Requests (CAR) and Forward Action Requests (FAR), plus minor issues are raised in the CDM validation protocol checklist and detailed in a separate form using Table 3 (Appendix C). In this form, note is made of actions taken by the Project Proponent to close outstanding CARs and respond to CLs and FARs:

Draft report corrective action, clarification, or forward action requests, or minor issues	Reference to CDM Validation Protocol Checklist	Summary of project participants' response	Final conclusion
List of CARs, CLs and FARs (and minor issues)	Reference to the validation protocol checklist question	Summary of response during the communication with the validation team	Summary of validation team responses and final conclusion.

Clarification Requests (CL): Where insufficient or unclear information is available and clarification or new information is required. A CL is raised specifying what additional information is required.

Corrective Action Requests (CAR): Where a non-conformance arises the Assessor shall raise a Corrective Action Request (CAR). A CAR is issued, where:

- Mistakes have been made that will influence the ability of the project activity to achieve real, measurable additional emission reductions;
- The CDM requirements have not been met; or
- There is a risk that emission reductions cannot be monitored or calculated.

The validation process may be halted until this information has been made available to the assessors' satisfaction. Failure to address a CL may result in a CAR. Information or clarifications provided as a result of a CL may also lead to a CAR.

Forward Action Requests (FAR): FARs shall be raised during validation to highlight issues related to project implementation that require review during the first verification of the project activity. FARs shall not relate to the CDM requirements for registration.

A 'Minor Issue' may be recorded for typographical errors or similar minor errors that do not have an impact on the compliance of the project to the CDM rules but nevertheless should be corrected to improve clarity.

2.5. Internal Quality Control

The process of validation and decision of the validation team has been subject to an independent Technical Review. The scope of the Technical Review process is to independently assess that all procedures have been followed, necessary requirements have been met, and all conclusions are justified. The final validation decision is based on the findings and conclusions of the validation team, assessing the compliance of the project activity with the CDM requirements, and the technical evaluation of the independent technical reviewer. The final report is then approved and signed off by the qualified signatory / final decision maker within ERM CVS

3. Validation findings

3.1. Main changes between the PDD version published for the global stakeholder comment period and the final version submitted for registration:

- The timeline of project development and the description of the project start date elaborated
- The accuracy of the meters has been addressed in the PDD
- The sensitivity analysis has been addressed in the PDD
- Other changes related to the CARs and CLs raised in the validation
- The crediting period has been updated in the PDD

3.2. Approval and Participation Requirements

The project participants are Hydrochina Guazhou Wind Power Co.,Ltd. authorised by P. R. China, and Macquarie Bank Limited authorised by United Kingdom. The host Party, P. R. China, and Annex I Party, United Kingdom have both ratified the Kyoto Protocol. Both Parties have established their respective Designated National Authorities (DNA) as per the participating requirements for CDM under the Kyoto Protocol.

The host country Letter of Approval (LoA) has been issued by the DNA of China, in August 2011 /24/. The LoA was received from the project participants and the authenticity of the LoA has been confirmed by cross-checking by ERM CVS against the list of approved projects on the web-site of the Chinese DNA. The letter confirms that:

- China ratified the Kyoto Protocol on 30th Aug 2002;
- Hydrochina Guazhou Wind Power Co.,Ltd. is authorised as a project participant by the PR China to voluntarily participate in the project activity;
- Guazhou Beidaqiao No.1 Wind Farm Project in Gansu Province, China complies with the permission requirements provided for in the Measures for Operation and Management of CDM projects in PR China, and assists PR China in achieving sustainable development;
- The LoA references the precise project title 'Guazhou Beidaqiao No.1 Wind Farm Project in Gansu Province, China'. This is consistent with the project title in the PDD.

The Annex 1 Party Letter of Approval (LoA) has been issued by the DNA of United Kingdom on 27 September 2011 /27/. The LoA was received from the project participants. The letter confirms that:

- United Kingdom ratified the Kyoto Protocol on 31 May 2002
- Macquarie Bank Limited is authorised as a project participant by United Kingdom to voluntarily participate in the project activity;
- Guazhou Beidaqiao No.1 Wind Farm Project in Gansu Province, China is voluntarily in the project
- The LoA references the precise project title 'Guazhou Beidaqiao No.1 Wind Farm Project in Gansu Province, China'. This is consistent with the project title in the PDD.

No information has been found that indicates that the project can be seen as a diversion of official development assistance (ODA) funding.

3.3. Project Design

Conformance of the PDD with EB guidelines

The PDD submitted for registration /1/ has been checked against the latest 'Guidelines for developing the Project Design Document' (version 7) /37/ and the latest template for the Project Design Document (version 3) /36/ available on the CDM website. It is confirmed that the final PDD is in compliance with the template and guidelines.

Conformance of the project design in the PDD with source documents such as the FSR

The project design and technical description of the project activity has been cross checked against the Feasibility Study Report /4/, the equipment purchase contract /19/, and also a physical site inspection conducted by the validation team to confirm that the description in the PDD reflects the proposed CDM project activity.

The total installed capacity of the project is 201 MW, consisting of 134 units with a unit capacity of 1.5MW. This is consistent with the Feasibility Study Report (FSR) which was developed by Northwest Hydro Consulting Engineers in January 2009 /4/, and approved by National Development and Reform Commission (NDRC) on 21 April 2009 /4/.

As per the FSR, the expected annual power delivered to the grid is 461,464 MWh. The power output will be delivered to the Gansu Provincial Grid which is part of Northwest China Power Grid (NWCPG) through an Anxi transmission line /4/. The load factor of this project is 26.2% /4/.

ERM CVS confirms that the description of the proposed CDM project activity as contained in the PDD sufficiently covers all relevant elements, is accurate, complete and provides the reader with a clear understanding of the nature of the proposed CDM project activity.

Annual power supplied to the grid was estimated by the third party design institute in the FSR, which was also approved by the local government. The net power supplied to the grid was calculated as 461,464MWh, based on 30 years of meteorological data in the area. The value has been checked by the validation team as reasonable and conservative, and in line with the FSR, and is there in compliance with the 'Guidelines for the reporting and validation of plant load factors' /7/.

Timeline and operational status of the project

A site visit was conducted by ERM CVS as described above, and the project was under construction at the time of the site visit.

The start date of the project activity is validated in section 3.5 below. The project timeline has been verified by checking all relevant documents such as construction contracts /20/ and the equipment purchase contract /19/.

The operational lifetime of the project activity is expected to be 22 years as stated in section C.1.2 of the PDD. This is consistent with the FSR and with the financial analysis, it is also consistent with wind turbine purchase contract /19/.

Permits and approvals:

ERM CVS has ensured that the project activity has all the relevant permits and approvals needed to be developed as a wind power project in the host country by means of review of the government approval to the project /4/.

Based on ERM CVS's local and sectoral knowledge, according to national procedures and regulations a project can only be approved when the necessary permits are in place.

Project location

As described in section A.4.1 of the PDD /1/, the project activity is located in Guazhou County, Gansu Province, in the People's Republic of China. The location described in the final PDD accurately reflects the location of the project, and the information is consistent with the FSR /4/, and has been confirmed by the site visit.

3.4. Baseline

Applicability of selected methodology

The project activity applies the valid and approved simplified baseline and monitoring methodology ACM0002 "Consolidated baseline methodology for grid-connected electricity generation from renewable sources", version 12.1.0 /6/. The methodology refers to the *Tool to calculate the emission factor for an electricity system* (Version 02) /9/ and the *Tool for the demonstration and assessment of additionality* (Version 05.2) /12/. The latest versions of the tools are applied.

ERM CVS has assured the applicability of the selected methodology to the proposed CDM project activity based on the following:

Table 1: Applicability conditions of the applied methodology

Applicability conditions	Justification in the PDD	Reference	Means of validation
The project activity is the installation, capacity addition, retrofit or replacement of a power plant/unit of one of the following types: hydro power plant/unit (either with a run-of-river reservoir or an accumulation reservoir), wind power plant/unit, geothermal power plant/unit, solar power plant/unit, wave power plant/unit or tidal power plant/unit;	The proposed project is the installation of a new wind power plant	/4/ SV	The information has been verified by ERM CVS through on site inspection, and checking the FSR and its approval by the government.
In the case of capacity additions, retrofits or replacements (except for wind, solar, wave or tidal power capacity addition projects which use Option 2: on page 10 to calculate the parameter $EG_{PJ,y}$): the existing plant started commercial operation prior to the start of a minimum historical reference period of five years, used for the calculation of baseline emissions and defined in the baseline emission section, and no capacity expansion or retrofit of the plant has been undertaken between the start of this	The proposed project is the installation of a new power plant at a site where no renewable power plant was operated prior to the implementation of the project activity;	/4/ SV	This has been confirmed by review of the FSR and site visit.

Applicability conditions	Justification in the PDD	Reference	Means of validation
minimum historical reference period and the implementation of the project activity;			
In case of hydro power plants, one of the following conditions must apply: ○ The project activity is implemented in an existing reservoir, with no change in the volume of reservoir; or ○ The project activity is implemented in an existing reservoir, where the volume of reservoir is increased and the power density of the project activity, as per definitions given in the Project Emissions section, is greater than 4 W/m²; or ○ The project activity results in new reservoirs and the power density of the power plant, as per definitions given in the Project Emissions section, is greater than 4 W/m²	Not applicable because the project activity is not a hydro power plant.	/1/ SV	Not applicable because the project activity is not a hydro power plant.
The methodology is not applicable to the following: • Project activities that involve switching from fossil fuels to renewable energy sources at the site of the project activity, since in this case the baseline may be the continued use of fossil fuels at the site; • Biomass fired power plants; • Hydro power plants that result in new reservoirs or in the increase in existing reservoirs where the power density of the power plant is less than 4 W/m².	The proposed project does not involve switching from fossil fuels to renewable energy sources at the site of the project activity;	/4/ SV	This has been confirmed by review of the FSR and site visit. Biomass and hydro are irrelevant to the project activity.

Project boundary

According to the methodology ACM0002 /6/, the spatial extent of the project boundary includes the project power plant and all power plants connected physically to the electricity system that the CDM project power plant is connected to. The information is correctly described in the PDD section B.3.

The validation of the project boundary has been performed by ERM CVS, through assessing the documentation such as the FSR and its approval /4/, and by the validation team's observations during the site visit. The whole power plant of the project to generate wind power and the connected regional grid are included in the project boundary. The DNA of the host country, NDRC of P. R. China, has published a delineation of the main electricity systems in China including the project electricity system of NWCPG /10/, and the information in the PDD is consistent with this.

No emissions other than those allowed by the methodology are expected to result from the project, and no greenhouse gas emissions occur within the proposed CDM project activity boundary, as a result of the implementation of the proposed CDM project activity, which are expected to contribute more than 1% of the overall expected average annual emissions reductions, which are not addressed by the applied methodology. This was confirmed by means of the on-site visit and by review of the detailed design of the project in the Feasibility Study Report /4/.

Baseline

As per VVM paragraph 105 /35/, no baseline alternative analysis is required if the approved methodology that is selected by the proposed CDM project activity prescribes the baseline scenario.

The methodology /6/ prescribes that if the project activity is the installation of a new grid-connected renewable power plant/unit, the baseline scenario is the following:

Electricity delivered to the grid by the project activity would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources, as reflected in the combined margin (CM) calculations described in the "Tool to calculate the emission factor for an electricity system" /10/. The calculation of baseline emissions is validated in detail in section 3.6.

The baseline of the project activity in the PDD correctly follows the methodology.

3.5. Monitoring Plan

As a newly constructed wind power project, the required monitoring parameters are the net electricity supplied by the project activity to the grid ($EG_{Facility, y}$) described in section B.7.1 of the PDD, the other parameters from the methodology are not relevant and do not need to be monitored for this type of project.

Requirements of the methodology for the parameter $EG_{Facility, y}$ are continuous measurement and at least monthly recording, and to cross check measurement results with records for sold electricity. As per the VVM paragraph 123, ERM CVS applied a two-step process to assessing compliance with the requirements of monitoring plan, and confirms the following:

- (a) the monitoring plan is fully in compliance with the requirements of the methodology ACM0002, version 12.1.0;
- (b) the monitoring arrangements described in the monitoring plan are feasible within the project design;
- (c) the project participants are able to implement the monitoring plan.

The assessment conducted by the DOE is by means of review of the documented procedures, interviews with relevant personnel, project plans and physical inspections of the proposed CDM project activity site. Further detail is given below.

Parameters monitored:

The following parameters are monitored:

$EG_{Facility, y}$	Quantity of net electricity generation supplied by the project plant/unit to the grid in year y
Title in line with methodology?	Yes
Data unit correctly expressed?	Yes
Parameter appropriately described?	Yes, the description is in line with the methodology
Source clearly referenced?	The source of data is the project activity site
Correct value provided for the purpose of PDD estimations?	Yes, the value has been checked against the FSR
Has this value been verified?	The value has been checked against the FSR
Measurement methods correctly described and in line with the methodology/tools?	In the PDD, the net quantity of electricity supplied by the project to the grid ($EG_{Facility, y}$) is monitored based on measurements of electricity output and electricity input. $EG_{Export, y}$ (annual electricity delivered to the grid by the proposed project), $EG_{Import1, y}$ (annual electricity imported from the grid by the proposed project) and $EG_{Import2, y}$ (annual electricity imported from the backup line by the proposed project) are measured and used as the basis for determining $EG_{Facility, y}$. This is considered a reasonable and transparent approach that provides for a conservative calculation of emission reductions, since electricity imported from the grid should be deducted from electricity delivered to the grid in order to determine the net quantity of electricity supplied by the project to the grid. $EG_{Facility, y} = EG_{Export, y} - EG_{Import, y}$ $EG_{Import, y} = EG_{Import1, y} + EG_{Import2, y}$
Correct reference to standards (i.e. for calibration and maintenance)?	The relevant national electric industry standards are specified /31/
Indication of accuracy provided?	N/A
QA/QC procedures described?	The PDD states that the electricity meters will be calibrated and checked annually, and readings will be double checked by records or sales receipts from the local grid company.
QA/QC procedures appropriate?	Yes, these procedures are in line with the methodology and are considered appropriate based on ERM CVS's local and sectoral knowledge.

$EG_{Export, y}$	Annual electricity delivered to the grid by the proposed project
Title in line with methodology?	This is an additional parameter added to improve the transparency of the monitoring plan, given that electricity input and output will be monitored separately.
Data unit correctly expressed?	Yes
Parameter appropriately described?	Yes, the description is appropriate
Source clearly referenced?	The source of data is electricity meters
Correct value provided for the purpose of PDD estimations?	Yes, the value has been checked against the FSR
Has this value been verified?	The value has been checked against the FSR

Measurement methods correctly described and in line with the methodology/tools?	Yes, the parameter will be continuously measured and monthly recorded. This is in line with the methodology requirements.
Correct reference to standards (i.e. for calibration and maintenance)?	The relevant national electric industry standards are specified /31/
Indication of accuracy provided?	Yes, accuracy class of the meters is specified as 0.2s
QA/QC procedures described?	The PDD states that the electricity meters will be calibrated and checked annually, and readings will be double checked by records or sales receipts from the local grid company.
QA/QC procedures appropriate?	Yes, these procedures are in line with the methodology and are considered appropriate based on ERM CVS's local and sectoral knowledge.

EG_{import1, y}	Annual electricity imported from the grid through the main line by the proposed project
Title in line with methodology?	This is an additional parameter added to improve the transparency of the monitoring plan, given that electricity input and output will be monitored separately.
Data unit correctly expressed?	Yes
Parameter appropriately described?	Yes, the description is appropriate
Source clearly referenced?	The source of data is electricity meters
Correct value provided for the purpose of PDD estimations?	Yes, the value has been checked against the FSR
Has this value been verified?	The value has been checked against the FSR
Measurement methods correctly described and in line with the methodology/tools?	Yes, the parameter will be continuously measured and monthly recorded. This is in line with the methodology requirements.
Correct reference to standards (i.e. for calibration and maintenance)?	The relevant national electric industry standards are specified /31/
Indication of accuracy provided?	Yes, accuracy class of the meters is specified as 0.2s
QA/QC procedures described?	The PDD states that the electricity meters will be calibrated and checked annually, and readings will be double checked by records or sales receipts from the local grid company.
QA/QC procedures appropriate?	Yes, these procedures are in line with the methodology and are considered appropriate based on ERM CVS's local and sectoral knowledge.

EG_{import2, y}	Annual electricity imported from the backup line by the proposed project
Title in line with methodology?	This is an additional parameter added to improve the transparency of the monitoring plan, given that electricity input and output will be monitored separately.
Data unit correctly expressed?	Yes
Parameter appropriately described?	Yes, the description is appropriate
Source clearly referenced?	The source of data is an electricity meter
Correct value provided for the purpose of PDD estimations?	Yes, the value has been checked against the FSR
Has this value been verified?	The value has been checked against the FSR
Measurement methods correctly described and in line with the methodology/tools?	Yes, the parameter will be continuously measured and monthly recorded. This is in line with the methodology requirements.
Correct reference to standards (i.e. for calibration and maintenance)?	The relevant national electric industry standards are specified /31/
Indication of accuracy provided?	Yes, accuracy class of the meters is specified as 0.5
QA/QC procedures described?	The PDD states that the electricity meters will be calibrated and checked annually, and readings will be double checked by records or sales receipts from the local grid company.
QA/QC procedures appropriate?	Yes, these procedures are in line with the methodology and are considered appropriate based on ERM CVS's local and sectoral knowledge.

Equipment:

The net electricity supplied by the project activity to the grid will be measured by bi-directional electricity energy meters (main meter M₁₁ and back-up M₁₂) with accuracy class of 0.2s at the main line and M₂₁ at the back-up line with the accuracy class of

0.5. The accuracy for all electricity meters is in compliance with the host country industrial standard *Technical Administrative Code of Electric Energy Metering* (DL/T448-2000) /31/.

This equipment setup is considered sufficient to carry out the monitoring requirements of the methodology, and the appropriate national standards have been followed. This has been cross checked against the standard DL/T448-2000 /31/.

Data:

The monitoring plan described in section B.7.2 of the PDD states that the project owner will record the readings of the meters monthly and all monitoring data and records will be kept at least for 2 years after the end of the last crediting period or the last issuance of CERs, whichever occurs later.

The data management procedures are considered appropriate to fulfil the monitoring requirements of the methodology and to ensure that emission reductions can be verified.

Organisation:

A figure is presented in the PDD to illustrate the monitoring organisation structure which will be carried out by the project developer during the project operation. An office manager will be appointed to be in overall charge of all relevant matters with the monitoring activity, which will include two aspects, monitoring and audit, in order to make sure the monitoring plan will be implemented efficiently and accurately. The organisational structure is considered sufficient to fulfil the monitoring requirements of the methodology and to ensure that emission reductions can be verified.

Quality Assurance and Quality Control:

The PDD describes calibration and trouble-shooting procedures in the monitoring plan. All electricity meters will be calibrated annually, and the reading from the back-up meter will be applied once the main meter fails. The net monthly electricity supplied to the Grid will be cross-checked against sale receipts issued by the grid company. In addition, internal auditing will be performed to assess the effectiveness of monitoring process and results.

The PDD contains sufficient description on how quality will be controlled and assured in the monitoring of emission reductions.

Feasibility of the monitoring plan:

Based on ERM CVS's local and sectoral knowledge, the measurement methods, recording procedures, meter maintenance and trouble-shooting procedures described in the monitoring plan are also required by the grid company when dealing with invoicing, and this will be specified in the PPA once signed. Therefore the procedures are considered standard and feasible in China and can fully meet the requirements of the CDM methodology. No difficulties are anticipated in implementing the operational and management structure or the monitoring plan as a whole.

3.6. Additionality

Project start date

The start date of the project activity is 24 October 2009, which is the date of the EPC contract (Engineering, Procurement and Construction contract) /20/, since this is the earlier contract signed by the project owner among the equipment purchase contracts /19/ and EPC contract /20/, therefore it is the earliest of the dates on which the implementation or construction or real action of the project activity has begun. All the documents have been checked and verified by ERM CVS.

The start date and the timeline of project implementation described in section B.5 of the PDD was also confirmed during the on-site interviews with the project developer.

ERM CVS confirms that the start date of the project activity, reported in the PDD, is in accordance with the "Glossary of CDM terms".

Table 2: Assessment to project start date determination

Activity	Date	Document / Evidence	Reference	Assessment
EPC contract (Engineering, Procurement and Construction contract) signed by the project developer (starting date of the project activity)	24 October 2009	EPC contract (Engineering, Procurement and Construction contract)	/20/	The signed contract has been checked, date and signatures are confirmed.
Wind turbine equipment purchase contract signed by the project developer	01 December 2009	Wind turbine equipment purchase contract	/19/	The signed contract has been checked, date and stamp are confirmed.

Prior consideration of the CDM and timeline of real and continuing actions to secure CDM status

The project activity is a new project with a starting date after 02 August 2008.

According to the 'Guidelines on the demonstration and assessment of prior consideration of the CDM' /38/, ERM CVS has validated that the CDM was seriously considered in the decision to implement the project activity.

The project participant informed the Host Party DNA /22/ and the UNFCCC secretariat /23/ in writing of the commencement of the project activity and of their intention to seek CDM status. Two notifications were made within six months of the project activity start date and contain the precise geographical location and a brief description of the proposed project activity, using the standardized form (F-CDM-Prior Consideration and China DNA form).

Confirmation from the UNFCCC secretariat that such a notification had been provided is available on UNFCCC website, and confirmation from the DNA of China has been checked /22/.

Identification of alternatives:

According to the applied methodology ACM0002 version 12.1.0 /6/, if the project activity is the installation of a new grid-connected renewable power plant/unit, the baseline scenario is the following:

Electricity delivered to the grid by the project activity would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources, as reflected in the combined margin (CM) calculations described in the "*Tool to calculate the emission factor for an electricity system*".

As per paragraph 105 of the VVM /35/, no analysis of baseline alternatives is required if the approved methodology that is selected by the proposed CDM project activity prescribes the baseline scenario.

The project activity is a wind power project that provides electricity to the local grid network which is dominated by fossil fuel fired power plants, and the methodology prescribes the baseline emissions as the product of the electrical energy baseline, expressed in kWh of electricity produced by the renewable generating unit multiplied by an emission factor.

Assessment of alternative baseline scenarios is therefore not required in the PDD.

Additionality determination:

According to the applied methodology ACM0002 version 12.1.0 /6/, the additionality of the project activity shall be demonstrated and assessed using the latest version of the "*Tool for the demonstration and assessment of additionality*" /12/.

Benchmark analysis is applied by the project, as per Step 2b of the "*Tool for the demonstration and assessment of additionality*".

The financial analysis was assessed by the validation team, including assessment of the spreadsheet and evidences relating to the input values to the financial analysis. The analysis was also assessed by referring to the 'Guidelines on the assessment of investment analysis' (version 04) /13/ by a financial expert assigned by ERM CVS, who has specific expertise in the assessment of financial analysis for CDM projects.

Investment analysis:

Determine appropriate analysis method

The PDD adopts the benchmark analysis (option III), which is appropriate, because there are other economic benefits associated with the project activity in addition to CER revenues (i.e. from selling electricity), hence option I is not applicable. Option II is also not applicable as the status-quo scenario of the project activity, continuation of electricity supply by the grid, does not represent a comparable investment opportunity to the proposed project. The alternative to the project is to 'do nothing' i.e. the project developer would make no investment. Therefore investment comparison analysis cannot be carried out.

Determination of the benchmark

According to the Interim Rules on Economic Assessment of Electrical Engineering Retrofit Projects /11/, the benchmark IRR for wind projects in China is regulated as 8% of the total investment (for post-tax project IRR). The benchmark is relevant for the project activity given that the benchmark is for power sector in China, and the benchmark is valid and commonly applied in the power industry in China.

This benchmark is considered appropriate as this is the common practice in the host country, based on ERM CVS's local and financial expertise. This is also in line with the financial indicator (post-tax project IRR) selected.

As per the VVM paragraph 112 /35/, ERM CVS has:

- (a) Determined that the type of benchmark applied (post-tax project IRR) is suitable for the type of financial indicator presented in the PDD: A post-tax project IRR is calculated which is consistent with the benchmark
- (b) Ensured that any risk premiums applied in determining the benchmark reflect the risks associated with the project type: considering the benchmark is regulated for power sector by the government and no additional risk premiums have been added to the benchmark, this is considered appropriate;
- (c) Determined that it is reasonable to assume that no investment would be made at a rate of return lower than the benchmark: given the benchmark is determined by the government taking into consideration circumstances in the country and sector, it is reasonable to conclude that the investment would not be made at a rate of return lower than the benchmark. This benchmark is widely applied in China according to the local and sectoral knowledge of the validation team.

Based on ERM CVS's local, sectoral and financial expertise, ERM CVS confirms the suitability of the benchmark applied in the investment analysis.

Assessment of parameters and assumptions

As per VVM paragraph 113 /35/ the project participants rely on values from a Feasibility Study Report (FSR) that is approved by the appropriate national authorities for proposed CDM project activities, and ERM CVS has ensured that:

- The FSR has been the basis of the decision to proceed with the investment in the project, i.e. that the period of time between the FSR (21 April 2009) /4/ and the investment decision (24 October 2009) /20/ is sufficiently short for ERM CVS to confirm that it is unlikely in the context of the underlying project activity that the input values would have materially changed;
- The values used in the PDD and associated annexes are fully consistent with the FSR /4/;
- On the basis of its specific local and sectoral expertise, confirmation is provided, by cross-checking or other appropriate manner, that the input values from the FSR are valid and applicable at the time of the investment decision. Further details on the cross checks carried out on the input parameters are given in the table below.

Table 3: Assessment of input values used in financial analysis

The total installed capacity of the proposed project is 201 MW. 5 previous windfarm CDM projects have already been registered with the same installed capacity in the same province (Reference numbers 3241, 3512, 4138, 4253 and 4278). Therefore ERM CVS has made a comparison of key input values against other registered wind projects in Gansu Province with the same installed capacity (201 MW). The data source is from the UNEP Risoe CDM pipeline /34/ and the PDDs and validation reports published on the website of the UNFCCC /33/.

Parameter	Reference	Validity at the time of investment decision and cross check of parameter
Total static investment	/4/ /19/ /20/ /33/ /34/	The value is sourced from the third party approved FSR, and the unit cost of the proposed project is calculated as 9,493 RMB/kW. ERM CVS has confirmed the information against the FSR /4/. According to the UNEP Risoe CDM pipeline /34/ and the website of UNFCCC /33/, the unit cost of other wind power projects with the same installed capacity in the same region (Gansu province) varies from 9,461 RMB/kW to 9,851 RMB/kW. And the average level of unit cost is 9,571 RMB/kW. The total investment cost of the proposed project is within the range of similar projects in the region, and it is slightly lower than the average level. In order to further cross check and confirm the investment costs, the principal investment related third party contracts were reviewed /19//20/. The total contracted value is approximately 1,840 million RMB. ERM CVS was able to review the cost breakdown of the same components in the FSR and the overall variance between the contracts and the FSR estimation of the cost is 3.56%, which shows that the estimation in the FSR is reasonable given that the project was still under construction and further expenditure was still yet to be incurred. Therefore the estimated value in the FSR is considered reasonable.
Annual power delivered to the grid	/4/ /33/ /34/	The annual power delivered to the grid is 461,464 MWh which is calculated by the third party design institute in the FSR and approved by the National government /4/. The meteorological data was monitored by Guazhou station with year from 1978-2007 /4/. ERM confirm that the annual power is reasonable with long time-series monitoring. ERM CVS has cross checked the annual power delivered to the grid against previously registered projects of a similar size in the province. Based on information in the UNEP Risoe CDM pipeline /34/ and the website of the UNFCCC /33/, there are 5 registered projects with the same installed capacity as the proposed project (201 MW) in the province, therefore these have been used as the basis for the comparison since they represent projects that are highly similar. Based on the data reviewed /33//34/, the annual power delivered to the grid is higher than the average of previously registered projects with the same installed capacity in the same province, which is 445,692 MWh According to <i>Guideline for the reporting and validation of plant load factors</i> /7/, the plant load factor defined ex-ante in the CDM-PDD has been verified by ERM CVS based on the following criteria: (a) The plant load factor was determined by a third party design institute which was contracted by the project participants, and the engineering estimation is based on 30 years of meteorological data in the area and one year of on site measurement /4/; (b) The plant load factor (26.2%) has been provided to the government while applying the project activity for implementation approval /4/. In summary, it is confirmed by ERM CVS that the plant load factor is valid and applicable at the time of investment analysis.
Electricity tariff	/4/	The tariff stated in the FSR is RMB 0.5206 Yuan/kWh (Inc. VAT). According to

	/33/ /34/	the FSR approval issued by NDRC on 21 April 2009 /4/, the tariff for the proposed project is the same, indicating that the tariff remains valid and applicable at the time of investment decision. If the highest tariff 0.585 RMB/kWh (incl. VAT) in Gansu province, as applied by the CDM Executive Board, is used in the IRR calculation sheet, the IRR is 7.64% which remains lower than the benchmark of 8%. Therefore, the tariff of 0.5206 RMB/kWh is considered reasonable. The detailed evaluation of the tariff can be found below.
Annual O&M cost	/4/ /33/ /34/	The project was under construction at the time of validation and therefore no invoices or receipts were available to cross check the O&M costs. Therefore in order to cross check the costs against third party price index data, the annual average O&M costs of the project were compared against other similar registered projects of the same installed capacity in the region. These projects were identified according to UNEP Risoe CDM pipeline /34/ and website of UNFCCC /33/. The Unit O&M costs (average annual O&M costs in RMB per MWh electricity delivered to the grid) of this project are 115 RMB/MWh on average, which are slightly higher than the average of 92.9 RMB/MWh for previous projects. This represents a difference of 23.7% above the average level of the 5 previously registered projects. However as discussed in the sensitivity analysis section below, Annual O&M cost would need to decrease by more than 62.5% in order for the IRR to exceed the benchmark. ERM CVS has reviewed the China Statistical Yearbook /21/, and can confirm that this is very unlikely to happen considering the labour and material costs have shown a steadily increasing trend with inflation and China's economic growth in recent decades.
Assessment period	/4/ /19/	The value is sourced from the FSR. It contains 22 years of operation lifetime and 3 years of construction period of the project, which is consistent with the PDD. The project lifetime is consistent with the technical lifetime of the wind turbines /19/ and therefore is determined to be reasonable.
Income tax rate	/4/ /17/	The tax rate is in line with the <i>Enterprise Income Tax Law of the People's Republic of China</i> (promulgated in the Document Order of the President of the People's Republic of China (No. 63) on March 16, 2007) which is effective since 01 January 2008. The FSR of the proposed project was approved in April 2009 and the investment decision was made in October 2009. Therefore the income tax rate /17/ is applicable at the time of the FSR writing and the investment decision.
Value added tax (VAT) rate	/4/ /28/	The value is derived from the national policy <i>Interim Regulation of the People's Republic of China on Value Added Tax</i> issued by the State Council on 13 December 1993 and revised in November 2008 /28/. The VAT rate is standard in China.
City maintenance & construction tax	/4/ /30/	The value is derived from the <i>Interim Regulations on City Maintenance and Construction Tax of the People's Republic of China</i> (promulgated in Document [1985] No. 19 of the State Council on Feb. 8, 1985) /30/. The tax rate is mandatory in China and applicable since 1985.
Educational surtax	/4/ /29/	The value is derived from the document <i>Decision of the State Council on Amending the Interim Provisions on the Collection of Educational Surcharges</i> (promulgated in the Document Order of the State Council No. 448 on August 20, 2005) which is effective since 2005 /29/. The tax rate is mandatory in China and applicable since 2005
Depreciation rate	/4/ /17/	The value is calculated based on the residual value (3%) which is in line with the <i>Enterprise Income Tax Law of the People's Republic of China</i> which is effective since 01 January 2008 /17/. The rate is evaluated as integrated and applicable.
Residual value	/4/ /17/	The accounting of residual value is in line with the <i>Enterprise Income Tax Law of the People's Republic of China</i> which is effective since January 1, 2008 /17/. The residual value of the project activity assets has been defined by the independent third party design institute in the approved FSR and is appropriately included as a cash inflow in the final year at the end of the assessment period. The level of residual value is considered appropriate based on ERM CVS's local and financial expertise.
Loan interest / amount	/4/ /41/	The interest rate derived from the FSR is in line with the benchmark interest rate regulated by the central bank of China (The People's Bank of China) since December 2008 /41/, which is applicable at the time of FSR preparation. The loan commitment amount for the project is no more than 1.6 billion RMB according to the loan commitment /43/. This is in line with the financing plan in the FSR. ERM CVS confirm that the loan interest / amount is valid and applicable at the time of investment analysis.

Calculation and comparison of financial indicators

The PDD presents the key input parameters and results of the IRR of the project, and ERM CVS has assessed the correctness of computations carried out by the project participants by reproducing the results using the IRR calculation spreadsheet /3/.

The validation team confirms that the calculations are correct, traceable, and consistent with the results of the FSR.

All input values used in the spreadsheet are consistent with the PDD and the FSR, and the calculation is in line with the *Guidelines on the Assessment of Investment Analysis* /13/, and is considered reasonable on the basis of ERM CVS's local and sectoral expertise and financial knowledge.

The project IRR calculated in the PDD and the spreadsheet is in line with the results of the FSR. The IRR of the project without CDM income (5.81%) is well below the benchmark of 8%, and hence it can be concluded that the project is additional.

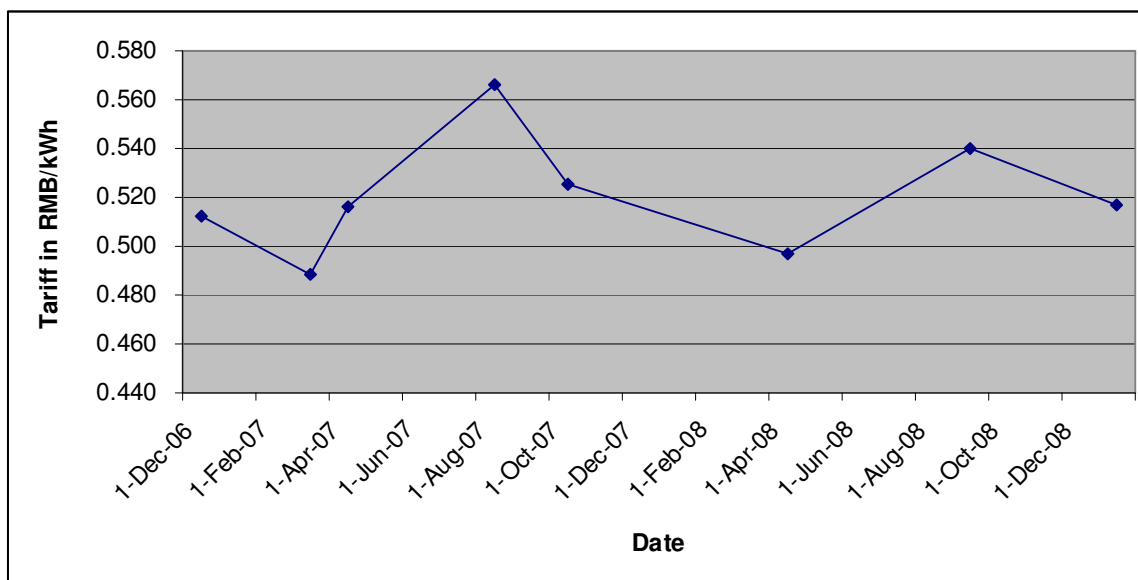
Electricity tariff:

The electricity tariff in the PDD and financial analysis is the same as that used in the FSR. The power tariff for the project is considered to be appropriate and applicable. The tariff has been cross checked against the project approval issued by NDRC in 21 April 2009 /4/, the tariff for the project was set as 0.5206 RMB/kWh, which is the same as the tariff applied in the proposed project activity.

In addition, ERM CVS has undertaken an examination of previous tariffs available in the province, which indicates that there has been no overall declining trend in wind power tariffs in the region since the first CDM wind projects in the province were registered, which further demonstrates that the tariff selected for the proposed project is appropriate.

The following graph illustrates the electricity tariff (including VAT) for previously registered wind power projects in Gansu province. The x (horizontal) axis refers to the starting date of the project activity, and the Y (vertical) axis refers to the electricity tariff cited in the registered documentation of each project. The data is taken from the IGES financial analysis database, which is based on the documentation on registered projects available on the CDM website. The projects where data was available represent 8 projects. The maximum tariff, including VAT, of the previously registered CDM projects in the region was 0.5658 RMB per kWh (according to the EB list the highest tariff in the province is 0.585 but the IGES data on considers tariffs adopted in CDM projects and this tariff was not actually awarded to a CDM project, but instead to a project titled 'Gansu Jieyuan Yumen Wind farm' which is not a registered CDM project) and the minimum, including VAT, was 0.4885 RMB/kWh. The average tariff was approximately 0.52 RMB/kWh, which is equal to the tariff applied in the investment analysis for the proposed project activity.

Graph 1: Electricity tariff (inc VAT) of previously registered projects in the province, plotted against the starting date of the project activity, illustrating the trend in electricity tariff over time.



Based on the analysis undertaken, ERM CVS has concluded that there is no overall downward trend in electricity tariffs in the province based on the information available. In addition, information relating to the highest tariff for Gansu Jieyuan Yumen Wind Farm Phase 1 was considered based on the 'Information note on the highest tariffs applied by the Executive Board in its decisions on registration of projects in the People's Republic of China'. The project is still additional even if the highest tariff (0.585 RMB/kWh) is applied to calculate the project IRR.

Therefore ERM CVS has concluded that the electricity tariff has been substantiated.

Sensitivity Analysis

A sensitivity analysis has been carried out to demonstrate the impact on the IRR of variations in the key input values to the financial analysis in accordance with the *Guidelines on the assessment of investment analysis /13/*. All costs and revenues greater than or equal to 20% of total costs / revenues have been included in the analysis. The variation in each parameter needed in order for the IRR to reach the benchmark, and the likelihood of such variations taking place, are explained. The analysis is assessed as appropriate.

Static investment

According to local knowledge and sectoral expertise, it is not likely that investment costs will reduce to the extent needed for the IRR to reach the benchmark. Key contracts including the equipment purchase contracts for wind turbines /19/ and The EPC contract (Engineering, Procurement and Construction contract) of Guazhou Beidaqiao No.1 Wind Farm Project /20/ signed by the project owner have been reviewed by ERM CVS and the variance between the contracted values and the FSR estimation of the costs of the same components is 3.56%. This shows that it is not likely for total investment to reduce by as much as 16.8% so that the IRR can cross the benchmark.

Annual generation

Annual generation would need to increase by more than 14.9% for the whole lifetime of 22 years in order for the IRR to exceed the benchmark.

It is confirmed by ERM CVS that this is very unlikely to happen considering that the technology is mature and that the operating hours are determined by an independent third party design institute based on a long time-series (30 years) of meteorological data /4/.

On-grid tariff

The tariff needs to increase by more than 14.9% for the whole lifetime so that the IRR can cross the benchmark. Based on the local and sectoral knowledge of the validation team, the tariff is not likely to change significantly under the policy framework because electricity tariffs are tightly regulated by the government and subject to price controls. Any future changes in the tariff are difficult to predict at the time of investment decision and based on ERM CVS's financial expertise and local and sectoral knowledge it is not reasonable to expect a rational investor to invest on the hope or assumption of a 14.9% increase in electricity tariff.

According to the highest tariffs applied by the EB which was a publication reflecting the report of EB 54, Paragraph 53, the highest tariff of wind power projects in Gansu province is 0.585 RMB/kWh (VAT incl.). Even if the highest tariff is applied in the financial analysis of the project, the IRR is calculated as 7.64% and still well below the benchmark.

In conclusion, the applied electricity tariff is considered to be valid and applicable at the time of investment decision, and it is not likely to be a factor that would lead the IRR to cross the benchmark.

Annual O&M cost

Annual O&M cost would need to decrease by more than 62.5% in order for the IRR to exceed the benchmark. ERM CVS has reviewed the China Statistical Yearbook /21/, and can confirm that this is very unlikely to happen considering the labour and material costs have shown a steadily increasing trend with inflation and China's economic growth in recent decades.

ERM CVS has reviewed the calculations for the sensitivity analysis which are presented in the IRR Spreadsheet /3/, and the computations are reproduced as correct and consistent with the information presented in the PDD.

Common Practice Analysis

A common practice analysis has been carried out in accordance with the *Tool for the demonstration and assessment of additionality /12/*. Projects are defined as similar to the project activity if they are wind projects in Gansu province and with an installed capacity between 100MW-300MW. Those are considered justified based on the following considerations:

- Provinces in China are very large in terms of geographical area, population size and natural resource availability. Investment and regulatory environment vary significantly between provinces in China;
- $\pm 50\%$ of the project scale is considered to be similar in terms of scale, based on ERM CVS's local and sectoral knowledge, and the range of projects selected for comparison (100MW-300MW) is therefore considered reasonable.

Based on 2010 China Wind Power Development Report /32/ and with reference to the CDM project documentation available on the UNFCCC website /33/, it is confirmed that only one project is identified as similar to the project activity: Gansu Anxi (Jiuquan phase 1) 100.5MW Wind Farm Project, that is operational and is not applying for CDM.

Since the similar project is identified, essential distinctions between it and the proposed project activity are explained in the PDD. According to the publicly accessible online database /32/, the Gansu Anxi (Jiuquan phase 1) 100.5MW Wind Farm Project has been successfully registered as a gold standard project, having demonstrated that it would not be built in the absence of carbon credit revenues. Therefore, essential distinctions have been demonstrated between the proposed project and the previous project, since the Gansu Anxi (Jiuquan phase 1) 100.5MW Wind Farm has access to additional revenues from the sale of Gold Standard emission reductions.

In sum, only one similar project was observed in the absence of the CDM, and this project was only developed due to the additional incentives provided by the Gold Standard. The fact that the previous project is able to source an additional revenue stream from the sale of Gold Standard credits is considered to constitute an essential distinction with the project activity (if the project activity was developed without revenues from the CDM), and therefore the proposed project activity is not considered to be common practice within the region.

Additionality summary

ERM CVS has assessed and verified the reliability and credibility of all data, rationales, assumptions, justifications and documentation provided by project participants to support the demonstration of additionality using its local knowledge and sectoral and financial expertise. ERM CVS can therefore conclude that the additionality of the proposed project activity has been demonstrated.

3.7. Calculation of GHG Emissions

ERM CVS has confirmed that the equations and parameters in the PDD have been correctly applied by comparing them to those in the selected approved methodology /6/. The GHG emission reductions of the project activity are calculated in accordance with the applied methodology: the baseline emissions are the product of the electrical energy baseline $EG_{Facility,y}$, expressed in kWh of electricity produced by the renewable generating unit multiplied by an emission factor. Project emissions are not considered according to the methodology ACM0002 /6/. Leakage is not to be considered if the energy generating equipment is not transferred from another activity.

Baseline emissions:

Grid emissions factor:

According to the requirement of ACM0002, the proposed project applied the steps of the *Tool to calculate the emission factor for an electricity system* /9/ to calculate the grid emission factor. The grid emission factor is calculated as a combined margin (CM) which is made up of the combination of operating margin (OM) and build margin (BM).

The DNA of the host country has published a delineation of the project electricity system and connected electricity systems, these delineations are used in the PDD as required by the *Tool to calculate the emission factor for an electricity system* /9/. The description in the PDD has been checked against the delineation published by China DNA /10/ as correct.

The emission factor of the grid is determined ex-ante for the 7 year crediting period following the tool to calculate the emission factor for an electricity system, version 2 (approved at EB 50) /9/. In accordance with the tool the weights of the operating margin and the build margin are 0.75:0.25.

The PDD was published on 30 November 2010, and the calculation of the grid emission factor is calculated based on the latest data available at the time of validation. The most recent three years of data at the time of PDD submission (2005-2007) are used based on the China Electric Power Yearbook /15/ and the China Energy Statistical Yearbook /14/ and applied in the NDRC guidelines /10/.

Operating Margin (OM): Method (a) (Simple OM) is used. This is appropriate since low-cost/must run resources constitute less than 50% of total grid generation in the average of the five most recent years (2003-2007). The Simple OM emission factor is calculated as the generation-weighted average emissions per electricity unit (tCO_2/MWh) of all generating sources serving the system, excluding low operating cost and must-run power plants. The data on fuel consumption and net electricity generation of each power plant /unit in the grid is not publicly available, therefore the simple OM is calculated based on data on the total net electricity generation of all power plants serving the system and the fuel types and total fuel consumption of the project electricity system (option B). Aggregated generation and fuel consumption data are used as more disaggregated data are not available. Low-cost/must run power resources in the grid include only renewable power generation, and the quantity of electricity supplied to the grid by these sources is known. Off-grid power plants are not included in the calculation. Therefore, Option B is appropriate for calculating the Simple OM emission factor. Net calorific values of each fuel type were obtained from the China Energy Statistical Yearbook, and IPCC 2006 default values /16/ were used for the oxidation factor and emission factor of each type of fossil fuel. The values used and the calculation of the simple OM is considered to be reasonable, and is in line with official data published by the Government of China /10/. The OM is calculated to be $1.0246 tCO_2 e/MWh$.

Build Margin (BM): Because plant specific fuel consumption and electricity generation data are not publicly available in China, the guidance given by the CDM Executive Board for a deviation from methodology AM0005 has been applied for calculation of the build margin (BM) emission factor for this project /39/. In accordance with this guidance, the build margin consists of the set of power capacity additions in the electricity system that comprises 20% of the generation capacity (in MW) of the system, that have been built most recently, based on the aggregate incrementally installed capacity of all generation sources in year y, and the aggregate incrementally installed capacity of all generation sources in year y-n, where n represents the number of years of historical data that need to be considered in order for the sample group to comprise 20% of the total system generation capacity (in MW). The emissions factor of fossil fuel fired power generation in the grid is calculated using the proportions of GHG emissions from solid, liquid and gaseous fuels in the total GHG emissions related to power generation as the weights, and the emission factors of the most advanced commercial generation technologies available in the host country (as published by the NDRC). Finally, the BM emission factor is calculated as the product of this emission factor of fossil fuel fired power generation and the proportion of fossil fuel fired power plants in the newly installed 20% capacity, based on data for years 2005-2007, contained in the power yearbooks 2006-2008 /15/. The BM is calculated as $0.6433 tCO_2 e/MWh$.

The OM and BM of the connected regional grid system (NWCPG) are calculated based on public and official data from the China Energy Statistical Yearbooks 2006-2008 /14/ editions and the China Electric Power Yearbooks 2006-2008 editions /15/. The weights ω_{OM} and ω_{BM} are selected as 0.75 and 0.25 respectively for wind projects according to the *Tool to calculate the emission factor for an electricity system* /9/. The combined margin is fixed ex-ante for the entire first crediting period.

Combined Margin (CM): The combined margin emissions factor is calculated as $0.75 * EF_{grid,OM,y} + 0.25 * EF_{grid,BM,y} = 0.92928$ tCO₂e/MWh. Therefore baseline emissions are calculated as $461,464 \text{ MWh} * 0.92928 \text{ tCO}_2\text{e/MWh} = 428,829 \text{ tCO}_2\text{e per annum}$.

The results are consistent with the data published by the DNA of China in the China Regional Grid Baseline Emission Factor in 2009 /10/.

Project emissions:

Project emissions are not considered for wind power project according to ACM0002 /6/. Therefore, project emissions are not considered in the ex-ante estimation of emission reductions.

Leakage:

According to the applied methodology /6/, leakage is not to be considered because the energy generating equipment is newly manufactured /19/ and is not transferred from another activity.

Conclusion:

As per the VVM paragraph 92 /35/, based on the information reviewed and calculations reproduced by the validation team, ERM CVS confirms the following:

- (a) All assumptions and data used by the project participants are listed in the PDD, including their references and sources;
- (b) All documentation used by project participants as the basis for assumptions and the sources of data are correctly quoted and interpreted in the PDD;
- (c) All values used in the PDD are considered reasonable in the context of the proposed CDM project activity;
- (d) The baseline methodology has been applied correctly to calculate project emissions, baseline emissions, leakage and emission reductions;
- (e) All estimates of the baseline emissions can be replicated using the data and parameter values provided in the PDD.

Parameters determined ex-ante

By reviewing the data and parameters used in the equations in accordance with the applied methodology ACM0002 /6/ and the Tool to calculate the emission factor for an electricity system, ERM CVS assessed that all data sources and assumptions of the data and parameters that will not be monitored and will remain fixed throughout the crediting period are appropriate and calculations are correct, applicable to the proposed CDM project activity and will result in a conservative estimate of the emission reductions.

To the data and parameters that will be monitored, ERM CVS confirmed that the estimates provided in the PDD for these data and parameters are reasonable including the value of data and references used. The parameters of $FC_{j,y}$, $NCV_{i,y}$, $EF_{CO_2,i,y}$, EG_y , $EF_{Coal,adv}$, $EF_{Oil,gas,adv}$, $CAP_{i,y}$ and Auxiliary Power Ratio have been discussed in the validation protocol. Please refer to the protocol section 8.15 for details.

3.8. Environmental and Sustainable Development Impacts

An Environmental Impact Assessment (EIA) Report has been performed according to Chinese regulations /5/. The EIA was conducted by Resource and Environment & Quality Evaluation Research Center of Lanzhou University in April 2008, and the document has been reviewed by ERM CVS. The EIA was approved, as per Chinese regulations, by the Gansu Environmental Protection Administration on 06 August 2008 /5/. The EIA report concludes that no significant environmental impacts are expected from the project. Impacts that are identified are subject to mitigation measures, and these are accurately described in the final PDD.

The letter of approval by the DNA of China /24/ confirms the contribution of the proposed CDM project activity to the sustainable development of the host Party.

3.9. Comments by Local Stakeholders

As per VVM paragraph 129 /35/, ERM CVS check Stakeholder Consultation Questionnaires /26/. The validation team confirm that:

- (a) Comments have been invited from local stakeholders that can reasonably be considered relevant for the proposed CDM project activity, through a questionnaire survey and consultation meeting;
- (b) The summary of the comments received as provided in the PDD is complete;
- (c) The stakeholders are all supportive to the project activity, no due account of any comments received should be undertaken.

To conclude, the local stakeholder consultation is adequate in the DOE's opinion.

3.10. Additional Findings

None

4. Conclusion and Validation Opinion

Name of Project	Guazhou Beidaqiao No.1 Wind Farm Project in Gansu Province, China
Basis of validation	ERM CVS based its validation work on: • CDM approved monitoring methodology ACM0002 v12.1.0 • Project Design Document Version 1.0 dated 25 November 2010 and the revised PDD Version 2.0 dated 25 February 2011 and revised PDD Version 3.0 dated 08 June 2011 • CDM Validation and Verification Manual v 1.2 (EB 55) • ERM CVS's internal CDM validation methodologies and protocols • CDM decisions and guidance issued by the CDM Executive Board • UNFCCC criteria for the Clean Development Mechanism • Host Country criteria for the Clean Development Mechanism
Responsibilities of ERM CVS	ERM CVS is responsible to provide a thorough independent third party assessment of the proposed CDM project activity to ensure that the proposed CDM project activity meets all the identified and applicable criteria for registration of projects under the CDM.
Responsibilities of Project Participants	Macquarie Bank Limited. and Hydrochina Guazhou Wind Power Co., Ltd. are responsible for preparing the PDD, supporting documentation and providing all necessary evidences to support the information included in the PDD.
Activities performed	ERM CVS conducted its activities in accordance with the CDM Validation and Verification Manual, approved v 1.2 (EB 55). The validation consisted of a review of project documentation, a site visit, interviews with relevant personnel, cross checking and ascertaining information through other reliable sources and on its sectoral, regional and local expertise and resolution of CLs and CARs pertaining to the project activity.
ERM CVS Conclusion	ERM Certification and Verification Services (ERM CVS) has performed the validation of the 'Guazhou Beidaqiao No.1 Wind Farm Project in Gansu Province, China' against the criteria for the Clean Development Mechanism as set out by the Conference of the Parties and the UNFCCC CDM Executive Board, and host country criteria. The validation employed standard auditing techniques, and a validation protocol checklist was used to carry out the validation. The project is a grid connected wind power project in Gansu Province, People's Republic of China. The Annex 1 Party for the proposed project activity is United Kingdom of Great Britain and Northern Ireland. The Party fulfils the criteria for participation in the CDM, and has issued a letter of approval for the proposed project and authorised the project participants. The LoA of the host Party, China, confirms the contribution of the proposed project towards sustainable development. The validation has provided sufficient evidence to demonstrate that the emission reductions would be additional to what would have taken place in the absence of the CDM project activity. The project meets the applicability criteria and correctly applies the approved methodology, and is therefore expected to result in real, measurable and long term reductions in greenhouse gas emissions. The monitoring plan provides for the collection and archiving of data sufficient to ensure that emission reductions can be verified. Nothing came to our attention to suggest that the project, if implemented as described, would not result in emission reductions of 428,829 tCO₂e per year on average over the first 7 year crediting period. In summary, it is the opinion of ERM CVS that the Project as described in the PDD Version 3.0 of 08 June 2011, meets all stated criteria of the CDM, correctly applies the methodology ACM0002 version 12.1.0, and is expected to result in real, measurable and long term emission reductions, and the DNA of the host Party has confirmed that the proposed project assists in meeting sustainable development criteria. ERM CVS therefore requests registration of the proposed project activity.
Signed on behalf of ERM CVS	
Name:	Melanie Eddis
Date:	28 September 2011

Appendix A: DOCUMENTS & INTERVIEWEES

DOCUMENT LIST

Reference	Document Title	Date
/1/	Project Design Document for Guazhou Beidaqiao No.1 Wind Farm Project in Gansu Province, China prepared by Beijing MD Energy Technology Co., Ltd	Version 1.0 of 25 November 2010 Version 2.0 of 25 February 2011 Version 3.0 of 08 June 2011
/2/	Emission Reduction Calculation spreadsheet prepared by MD Energy Technology Co., Ltd	Version 1.0 of 25 November 2010 Version 2.0 of 25 February 2011 Version 3.0 of 08 June 2011
/3/	IRR calculation spreadsheet prepared by MD Energy Technology Co., Ltd	Version 1.0 of 25 November 2010 Version 2.0 of 25 February 2011 Version 3.0 of 08 June 2011
/4/	The Feasibility Study Report (FSR) of Guazhou Beidaqiao No.1 Wind Farm Project in Gansu Province, China prepared by Northwest Hydro Consulting Engineers and the approval of FSR issued by NDRC	FSR prepared in January 2009 and approved on 21 April 2009
/5/	The EIA report of Guazhou Beidaqiao No.1 Wind Farm Project in Gansu Province, China prepared by Resource and Environment & Quality Evaluation Research Center of Lanzhou University and the approval of EIA issued by Gansu Environmental Protection Administration	Completed on April 2008 Approved on 06 August 2008
/6/	CDM Executive Board: ACM0002 "Consolidated methodology for grid-connected electricity generation from renewable sources", version 12.1.0	17 September 2010
/7/	CDM Executive Board: Guideline for the reporting and validation of plant load factors version 01.	11 July 2009
/8/	CDM Executive Board: Guideline on the demonstration and assessment of prior consideration of the CDM.	Version 01 of 2 August 2008 Version 02 of 17 July 2009
/9/	CDM Executive Board: Tool to calculate the emission factor for an electricity system (version 2)	16 October 2009
/10/	China NDRC, the emission factor calculation for each power grid of China, NDRC official website: http://ghs.ndrc.gov.cn/qfzjz/t20090703_289357.htm	02 July 2009
/11/	Interim Rules on Economic Assessment of Electrical Engineering Retrofit Projects. Beijing: China Electric Power Press.	2003
/12/	CDM Executive Board: Tool for the demonstration and assessment of additionality, version 5.2.	16 May 2008
/13/	CDM Executive Board: the Guidelines on the Assessment of Investment Analysis version 04	3 June 2011
/14/	China Energy Statistic Yearbook	2006-2008
/15/	China Electric Power Yearbook	2004-2008
/16/	IPCC: Revised 2006 IPCC Guidelines for National Greenhouse Gas Inventories Reference Manual.	2006
/17/	Enterprise Income Tax Law of the People's Republic of China	issued on 16 March 2007
/18/	The business license of Guazhou Beidaqiao No.1 Wind Farm Project in Gansu Province, China	18 August 2009
/19/	The purchase contract for wind turbines between Guazhou Beidaqiao No.1 Wind Farm Project in Gansu Province, China and Beijing Goldwind Science & Creation Windpower Equipment Co., Ltd and Sinovel Wind (Group) Co., Ltd.	01 December 2009
/20/	The EPC contract (Engineering, Procurement and Construction contract) of Guazhou Beidaqiao No.1 Wind Farm Project	24 October 2009
/21/	China Statistical Yearbook	2006-2009
/22/	The notification to DNA of China of the commencement of Guazhou Beidaqiao No.1 Wind Farm Project in Gansu Province, China and the intention to seek CDM status	Signed by Guazhou Beidaqiao No.1 Wind Farm Project in Gansu Province, China dated 25 March 2010 and signed by DNA of China dated 06 April 2010
/23/	The notification to EB of the commencement of Guazhou Beidaqiao No.1 Wind Farm Project in Gansu Province, China and the intention to seek CDM status	Submitted on 25 April 2010
/24/	The letter of approval by the DNA of China	August 2011
/25/	The CDM consultancy agreement between Guazhou Beidaqiao No.1 Wind Farm Project in Gansu Province, China and Beijing MD Energy Technology Co., Ltd	19 January 2010
/26/	50 copies of questionnaires for stakeholders consultation	28 April 2009
/27/	The letter of approval by the DNA of United Kingdom	27 September 2011

/28/	Interim Regulation of the People's Republic of China on Value Added Tax, http://www.gov.cn/zwqk/2008-11/14/content_1149516.htm	10 November 2008
/29/	Decision of the State Council on Amending the Interim Provisions on the Collection of Educational Surcharges, http://www.gov.cn/zwqk/2005-09/27/content_70440.htm	20 August 2005
/30/	Provisional Regulations of the People's Republic of China on City Maintenance and Construction Tax, http://www.gov.cn/banshi/2005-08/19/content_24817.htm	28 February 1985
/31/	The national industry standard of the Technical Administrative Code of Electric Energy Metering (DL/T448-2000)	01 January 2001
/32/	2010 China Wind Power Development Report issued by Li Junfeng, Shi Pengfei and Gaohu	October 2010
/33/	The website of United Nations Framework Convention on Climate Change http://cdm.unfccc.int/Projects/projsearch.html	24 January 2011
/34/	UNEP Risoe CDM pipeline http://cdmpipeline.org/publications/CDMpipeline.xlsx	24 January 2011
/35/	CDM Validation and Verification Manual (Version 1.2, EB 55)	30 July 2010
/36/	UNFCCC CDM PDD Form (CDM PDD), Version 03	28 July 2006
/37/	Guidelines for Completing the CDM-PDD and the CDM- NM (version 07)	02 August 2008
/38/	Guidelines on the demonstration and assessment of prior consideration of the CDM	11 September 2009
/39/	The guidance for deviation in use of methodology AM0005 by several project activities in China by EB. http://cdm.unfccc.int/Projects/Deviations	2005
/40/	The training material for Guazhou Beidaqiao No.1 Wind Farm Project in Gansu Province, China	20 October 2010
/41/	Historical benchmark interest rate published by The People's Bank of China http://www.pbc.gov.cn/detail.asp?col=462&ID=2480	December 2008
/42/	Economic Evaluation Method and Parameter of Construction Projects	2006
/43/	The loan contract between the project owner and the China Development Bank	21 July 2010
/44/	The development of wind turbine in China (http://www.newenergy.org.cn/html/0085/5290817781.html)	29 May 2008

List of Interviewees

Date	Name	Position	Subject Discussed
22 December 2010	Yan Xingguo	Vice President of Hydrochina Guazhou Wind Power Co.,Ltd	➤ Project background information. ➤ Project technology, operation, maintenance and monitoring capability. ➤ Project additionality. ➤ Project monitoring and management plan. ➤ Project approval status (incl. EIA approval, CDM project approval status)
	Cheng Zhengming	Engineering Manager of Hydrochina Guazhou Wind Power Co.,Ltd	

Date	Name	Position	Subject Discussed
	Ruth Wang	Project Manager of Macquarie Bank Limited	➤ Stakeholder consultation process ➤ Project approval status (incl. EIA approval, CDM project approval status) ➤ Stakeholder consultation process ➤ Environmental Impact ➤ Society and Sustainable Development
	Zhao Ruihong	Technical Director of Beijing MD Energy Technology Co., Ltd	➤ Applicability of selected methodology. ➤ Baseline determination. ➤ Emission reductions calculation. ➤ Emission reduction ➤ Monitoring plan
	Sun Yangyang	Project Manager of Beijing MD Energy Technology Co., Ltd	

Appendix B: CDM Validation Protocol Checklist

	Checklist Question	Reference	Comment	Draft Conclusion OK/CAR/CL	Final Conclusion OK/ NOT OK
1.	PDD Format (CDM VVM EB 55 para. 55 - 57)				
1.1	Is the PDD prepared in accordance with the latest template and guidance by the CDM EB? http://cdm.unfccc.int/Reference/PDDs_FORMS/PDDs/index.html	/1/	Yes, PDD follows the latest large scale template from the CDM website.	OK	OK
1.2	Does the language make sense and is it clear?	/1/	Yes, language does make sense and is clear except for CLs and CARs raised in the following	OK	OK
2.	Project Title	PDD A.1			
2.1	Does the used project title clearly enable identification of the unique CDM activity?	/1//4/	Yes. The title of the project activity is "Guazhou Beidaqiao No.1 Wind Farm Project in Gansu Province, China". The project title is based on the location where the Project is situated, enabling clear identification of the said CDM activity.	OK	OK
2.2	Is the version number and the date given? Is this consistent with the project's timeline?	/1//4/	Yes, PDD present version is 1.0, dated 25/11/2010. PDD version number will be updated if the PDD is amended based on draft validation findings.	OK	OK
3.	Project Description (CDM VVM 55 para. 58-64)	PDD A.2			
3.1	Does the PDD contain a clear description of the project activity, with regard to its nature and technical implementation? Does Section A2 include: a. A brief summary of the technology employed, b. A brief summary of the sources and gases included in the project boundary c. The PP's opinion regarding the contribution of the project to sustainable development	/1//4/	A clear description of the project is given with the exception of certain issues raised as CARs below. The description has been cross checked during the observations made during the site visit and also against the Feasibility Study Report (FSR). a. Yes, a brief summary of the applied technology is included. b. Yes, it mentions that the project activity will reduce CO ₂ emissions through displacing fossil fuel-dominated power generation of the Northwest China Power Grid (NWPg). c. Yes, the PP's opinion regarding contribution of the project towards sustainable development is included. The project description is in line with the FSR and other supporting documents. Also the project description has been verified against observations made on site.	OK	OK
3.2	Does the description deliver a transparent overview of the project activities and cover all relevant elements?	/1//4/	Yes. The project will install and operate 134 wind turbines with a total capacity of 201 MW which will supply an average annual generation of 461,464 MWh to Northwest China Power Grid (NWPg) and replace the same amount of electricity generation	OK	OK

Validation Report

	Checklist Question	Reference	Comment	Draft Conclusion OK/CAR/CL	Final Conclusion OK/ NOT OK
			by fossil fuel fired power plants connected to NWPG.		
3.3	Has a physical site inspection been undertaken to confirm that the description in the PDD reflects the proposed CDM project activity?	SV	Yes, a site inspection was undertaken from 21 to 23 December 2010. The description of the project is in line with site observations. In addition, the FSR was reviewed and is consistent with the project description contained in the PDD.	OK	OK
3.4	Does section A.2 also indicate the baseline situation, and the historical situation at the facility, if this is different to the baseline?	/1//4/	The baseline situation is clearly described as the continuation of power supply from the NWPG as per the applied methodology. Details of the grid, the purpose of the project and its contributions to GHG emissions reduction and contribution towards sustainable development are included in the PDD. The proposed CDM project activity does not involve the alteration of an existing installation or process and the differences to the pre-project situation are described. The project has not yet started, which was confirmed during the onsite inspection.	OK	OK
3.5	If the proposed CDM Project Activity takes place in an existing facility, installation or process, or modifies an existing facility, installation or process, is a complete and clear description of that facility, installation or process given?	/1//4/	The proposed CDM Project Activity does not take place in an existing facility, installation or process, or modify an existing facility, installation or process.	OK	OK
3.6	How has this description been validated?		N/A	N/A	N/A
3.7	Is the description of the existing facility, installation or process consistent with information provided in other parts of the PDD such as common practice and baseline selection?		N/A	N/A	N/A
3.8	Is all information provided in the project description consistent with information provided in later sections of the PDD?	DR	Yes, a desk review of the PDD has been undertaken; all information provided in the project description is consistent with information provided in later sections of the PDD.	OK	OK
4.	Technical Description	PDD A.4			
	Location of Project	PDD A.4.1			
4.1	Does the information provided on the location of the project activity allow for a clear identification of the site(s)? How was the site location confirmed? (e.g. site visit, planning documents)	SV /1//4//5/	Yes. The project is located at Guazhou County, Jiuquan City, Gansu Province, China, and this has been confirmed during the site visit made on 22 December 2010 as well as verifying the FSR.	OK	OK
	Category/ Sectoral Scope	PDD A.4.2			
4.2	Is the category (sectoral scope) of the project activity indicated and correct?	/1/	Yes. The project activity falls under sectoral scope 1: Energy industries (renewable/non-renewable sources), which is correct for a wind farm project, and is in accordance with the selected	OK	OK

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	Checklist Question	Reference	Comment	Draft Conclusion OK/CAR/CL	Final Conclusion OK/ NOT OK
			methodology ACM0002.		
	Technology to be Employed by the Project Activity	PDD A.4.3			
4.3	Is there a clear description of the baseline scenario, as identified in section B.4? This should include: a. An indicative list of the equipment(s) and systems that would have been in place in the absence of the project activity (if any) b. Information about the age and average lifetime of the baseline facility based on manufacturer's specifications and industry standards (if applicable) c. Installed capacities, load factors and efficiencies of the baseline facility (if applicable) d. An explanation of how the same types and levels of services provided by the project activity would have been provided in the baseline scenario.	SV	The PDD includes a clear description of the baseline scenario. a. Not applicable, since the project does not take place at an existing facility. In the absence of the project activity, electricity would continue to be supplied by the grid, details of which are given in the PDD. b. Not applicable, since there is no baseline facility. c. Not applicable, since there is no baseline facility. d. It is clearly described that the electricity would have been provided by the existing scenario before implementation of the project activity i.e. electricity demand is met by the existing generation mix grid that makes up the NWPG, as defined by the combined margin calculations detailed in the PDD.	OK	OK
4.4	If the scenario existing prior to the start of the implementation of the project activity is different from the selected baseline scenario, is there a clear description of the pre-existing scenario, with a list of the equipment(s) and systems in operation at that time?	/1//4/	Not applicable	N/A	N/A
4.5	Is the technology to be employed by the project activity clearly described and is it consistent with information provided elsewhere in the PDD? a. List of main technologies involved b. List of main equipment and installations c. The lifetime of the project equipment d. Monitoring equipment and its location e. Capacities, load factors and efficiencies f. The emissions sources and the greenhouse gases involved in the project activity g. Existing and forecast energy and mass flows and balances h. Interaction with processes/equipment outside the project boundary, if any, is stated.	SV, DR /1//4//6/	The PDD includes a description of the technology to be employed by the project, and this has been cross checked against the detailed design of the project in the FSR, and during the site visit. a. The main technologies involved are listed and the information has been verified against the background documents, especially FSR and was double checked by interviews with PPs. b. Main equipments are described and the information has been checked against the purchase contract for equipment, and was also verified during the site visit. c. Lifetime of the project equipments is provided as 22 years in the PDD and this is verified to be consistent with the FSR, also cross checked via review of equipment purchase contract. However, this information has not been included in section A.4.3 of the PDD, as required by the Guidelines for Completing the PDD. d. The monitoring equipments are not described in the PDD section A.4.3.	CL	OK

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	Checklist Question	Reference	Comment	Draft Conclusion OK/CAR/CL	Final Conclusion OK/ NOT OK
			e. The capacities were specified, however, load factor was not described in the PDD section A.4.3. f. In accordance with ACM0002, there are no emission sources involved in the project activity since this is a wind power project. However, this information and the GHGs and emission sources in the baseline scenario have not been included in section A.4.3 of the PDD, as required by the Guidelines for Completing the PDD. g. The electricity generation was specified, however, the load factor and the effective electricity coefficient were not described in the PDD section A.4.3. h. Not applicable CL 1: The following technical information is not provided in section A.4.3 of the PDD, as required by the PDD guideline: (1) the lifetime of the project equipment and the reference for the project operational lifetime should be added in section C.1.2; (2) monitoring equipments; (3) technology transfer; (4) load factor; and (5) GHGs emission sources. (6) effective electricity coefficient		
4.6	Does the description of the technology to be applied provide sufficient and transparent input/ information to evaluate its impact on the greenhouse gas balance?	/1//4/	Yes. It was clear except some information that was missing – please see CL 1 as mentioned above.	CL-1	OK
4.7	Does the implementation of the project activity require any technology transfer from Annex-1-countries to the host country(ies)?	/1//4/	No, all equipment will be domestically produced. This has been verified against the FSR.	OK	OK
4.8	Does the project use state of the art technology and / or does the technology result in a significantly better performance than any commonly used technologies in the host country? Is the technology implemented by the project activity environmentally safe?	/1//4//5//20/	Wind technology is mature in the host country, and the project adopts modern efficient wind turbine design; which was verified against the FSR. The project does not result in any emissions of GHGs; therefore represents clean technology in comparison with the overall generation mix operating in the grid (NWPG). The technology is wind technology, it is considered environmentally safe as indicated in the EIA report prepared by Gansu Environmental Science Research Institute.	OK	OK

Validation Report

	Checklist Question	Reference	Comment	Draft Conclusion OK/CAR/CL	Final Conclusion OK/ NOT OK
4.9	Is the project technology likely to be substituted by other or more efficient technologies within the project period?	/1//4/20/	The project adopts advanced domestically produced wind power technology, of a standard widely adopted in the host country. There is no indication that the project technology is likely to be substituted by other or more efficient technologies within the project period.	OK	OK
4.10	Does the project require extensive initial training and maintenance efforts in order to be carried out as scheduled during the project period? Is information available on the demand and requirements for training and maintenance?	SV, /I/	Yes. For this specific project, the project has not started operation. CL 2: Please provide training material for project operation.	CL-2	OK
4.11	Is a schedule available for the implementation of the project and are there any risks for delays?	SV, IV /1//4/	Yes, no risk for delay was identified based on site observation and interview.	OK	OK
	Public Funding	PDD A.4.5			
4.12	Is the information provided on public funding provided in compliance with the actual situation or planning as available by the project participants?	/1//4/	No public funding will be used for the project activity according to the interview with the project developer and the financial analysis in the FSR.	OK	OK
4.13	If the project involves public funding from an Annex 1 country, have the annex 1 parties involved provided an affirmation that such funding does not result in a diversion of official development assistance?		Not applicable	N/A	N/A
5.	Approval and Participation (CDM VVM EB 55 para.44 – 50 and para. 51 - 54)	PDD A.3			
5.1	Are project participants listed in tabular form in section A.3 of the PDD? Is this information consistent with the contact details provided in Annex 1 of the PDD and other project documentation (Letters of Approval and Modalities of Communication)?	/1/	Hydrochina Guazhou Wind Power Co.,Ltd. is listed as the host country (China) project participant. Macquarie Bank Limited is listed as the Annex I country (United Kingdom) project participant. The annex 1 of the PDD also gives information about the project participant, which are consistent.	OK	OK
5.2	Has the Host Party provided a Letter of Approval (LoA) with clear referencing and supporting documentation? Does the LoA confirm: - o Ratification of the Kyoto Protocol - o Voluntary Participation - o Contribution to Sustainable Development - o Reference to the precise project title in the PDD	/1/	CAR 1: The LoAs from host party and Annex I party have not been provided yet.	CAR-1	OK
5.3	Was the LoA received directly from the DNA or from the project participants? Has the Host Party LoA been issued	/1/	CAR 1: The LoAs from host party and Annex I party have not been provided yet.	CAR-1	OK

Validation Report

	Checklist Question	Reference	Comment	Draft Conclusion OK/CAR/CL	Final Conclusion OK/ NOT OK
	by the respective DNA? How has this been confirmed?				
5.4	Has the Annex I Party provided a Letter of Approval (LoA) with clear referencing and supporting documentation? Does the LoA confirm: - o Ratification of the Kyoto Protocol - o Voluntary Participation - o Contribution to Sustainable Development - o Reference to the precise project title in the PDD	/1/	CAR 1: The LoAs from host party and Annex I party have not been provided yet.	CAR 1	OK
5.5	Was the LoA received directly from the DNA or from the project participants? Has the Annex I Party LoA been issued by the respective DNA? How has this been confirmed?	/1/	CAR 1: The LoAs from host party and Annex I party have not been provided yet.	CAR 1	OK
5.6	If either LoA contains additional specification or conditions of the project activity, then has the request for registration been based on the documents specified in the LoA?	/1/	CAR 1: The LoAs from host party and Annex I party have not been provided yet.	CAR 1	OK
5.7	If the LoA references a specific version of the Validation Report and this version cannot be submitted, then has either of the following been submitted? a) a statement indicating final LoA has not been received - or b) an updated Validation Report	/1/	CAR 1: The LoAs from host party and Annex I party have not been provided yet.	CAR 1	OK
6.	Baseline and Monitoring Methodology (CDM VVM EB 55 (para. 65-92))	PDD B			
	Title and reference of the approved baseline and monitoring methodology?	PDD B.1			
6.1	Are the number, version and reference of the methodology clearly and correctly stated? Has the methodology been previously approved by the CDM Executive Board? http://cdm.unfccc.int/Reference/Procedures/Meth_proc02_v13.pdf	/1//6/	Yes. The GSP PDD adopts the methodology ACM0002 "Consolidated methodology for grid-connected electricity generation from renewable sources", version 12.1.0 of 17 September 2010, and it is valid at the time of validation submission. The PDD was uploaded for GSP on 30 November 2010.	OK	OK
6.2	Are the Tools applicable to the methodology correctly referenced, including the correct version number(s) valid at the time of registration submission?	/1//6/	Yes. The PDD references: • "Tool for the demonstration and assessment of additionality" (Version 05.2), approved at EB 39. • "Tool to calculate the emission factor for an electricity system" (Version 2), approved at EB 50. These are the most recent versions of the tools at the time of the	OK	OK

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	Checklist Question	Reference	Comment	Draft Conclusion OK/CAR/CL	Final Conclusion OK/ NOT OK
			validation submission.		
	Justification for the choice of methodology and why it is applicable	PDD B.2			
6.3	Have any sources of greenhouse gas emissions been identified by the DOE, within the project boundary following project implementation, which are expected to contribute more than 1% of the overall expected average annual emissions reductions, and which are not addressed by the applied methodology?	SV /1//4//6/	No. The project is a wind power project and there is no fossil-fuelled backup generator as part of the project. This has been verified against the FSR.	OK	OK
6.4	Is the methodology fully applicable to the proposed project? For each of the applicability criteria: a. Is the criterion discussed in the PDD? b. Is compliance provable? c. Is evidence provided in the PDD to prove applicability? - o Has compliance with the criterion been verified (by checking evidence provided, sector/ local knowledge etc)?	SV, I /1//4//6/	a. The applicability criteria of the applied methodology are justified in section B.2 of the PDD. The project meets all the applicability criteria stated as per methodology ACM0002 version 12.1.0. b. The following criteria are applicable for this project: • <i>The project activity is the installation or modification/retrofit of a power plant/unit of one of the following types: hydro power plant/unit (either with a run-of-river reservoir or an accumulation reservoir), wind power plant/unit, geothermal power plant/unit, solar power plant/unit, wave power plant/unit or tidal power plant/unit.</i> The proposed project is a new grid-connected zero-emission renewable power generation activity from wind source. This has been described in the FSR. • <i>The methodology does not apply to project activities that involve switching from fossil fuels to renewable energy sources at the site of the project activity, since in this case the baseline may be the continued use of fossil fuels at the site.</i> The proposed project does not involve an on-site switch from fossil fuels to a renewable source. This has been verified with the FSR and on site inspection. c. Yes. The evidences provided in the PDD are considered to be applicable. This has been checked against the FSR and verified during the site visit.	OK	OK

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	Checklist Question	Reference	Comment	Draft Conclusion OK/CAR/CL	Final Conclusion OK/ NOT OK
			d. Yes, The criteria have been checked against the FSR and verified during the site visit.		
6.5	Was there a request for clarification, revision or deviation made for the adopted methodology in relation to the proposed project activity? If so, were the correct procedures provided by the CDM EB followed?		Not applicable	N/A	N/A
	Description of sources and gases included in the project boundary	PDD B.3			
6.6	Does the PDD correctly describe the project boundary, including the physical delineation of the proposed CDM project activity, in compliance with the requirements of the selected baseline methodology, and is this consistent with site observations and other documentation provided?	/1//6//10/	The PDD correctly describes the project boundary. The electricity generated by the proposed project will be directed/supplied to the NWPG. The spatial extent of the project boundary includes the project site and all power plants connected physically to the NWPG. The areas covered by NWPG include. Gansu Province, Qinghai Province, Shaanxi Province, Ningxia Province and Xinjiang Autonomous Region. This has been verified using the definition of the grid boundary provided by the China DNA.	OK	OK
6.7	Baseline emissions: Have all sources and GHGs required by the methodology been included within the project boundary? For each potential source: a. Are source(s) and gases discussed by the PDD? b. Is inclusion / exclusion justified? c. Is explanation/ justification sufficient? d. Is the inclusion/ exclusion consistent with the monitoring plan?	/1/ /6/	Yes. The description of sources and GHGs is in accordance with ACM0002. For each potential source the following is confirmed: a. All sources and gases identified in the methodology are discussed in the PDD. b. Yes. CO ₂ emissions from electricity generated from fossil fuel fired power plants that are displaced due to the project are included. This inclusion is considered as justified. CH ₄ and N ₂ O emissions are excluded since these are minor emissions sources according to the methodology. The exclusion is considered as justified. c. Yes, the justification is sufficient. d. Yes, this is consistent with the provisions of the monitoring plan	OK	OK
6.8	Project emissions: Have all sources and GHGs required by the methodology been included within the project boundary? For each potential source: a. Are source(s) and gases discussed by the PDD? b. Is inclusion / exclusion justified? c. Is explanation/ justification sufficient?	/1/ /6/	a. All sources and gases identified in the methodology are discussed in the PDD. b. No sources of project emissions exist for a wind project according to the methodology. c. Yes. d. Yes, this is consistent with the provisions of the monitoring	OK	OK

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	Checklist Question	Reference	Comment	Draft Conclusion OK/CAR/CL	Final Conclusion OK/ NOT OK
	d. Is the inclusion/ exclusion consistent with the monitoring plan?		plan.		
6.9	If the methodology allows project participants to choose whether a source or gas is to be included within the project boundary, have the project participants sufficiently justified that choice? Is the justification reasonable, based on assessment of supporting documented evidence and corroborated by observations if required?		N/A	N/A	N/A
6.10	For large scale projects, is a diagram given to illustrate the project boundary, including all the key equipment, systems and flows of mass and energy, as well as the emissions sources and gases included in the project boundary?	/1//4//6/	The diagram is consistent with the FSR, however it does not include the flows of mass/energy and the monitoring variables. CL 3: Please include the flows of mass/energy and monitoring variables in the diagram in section B.3 of the PDD, as required by the PDD Guidelines.	CL 3	OK
6.11	Overall, is the identified boundary and the selected sources and gases justified for the project activity?	/1/	Yes, the identified boundary and the selected sources and gases are justified for the project activity.	OK	OK
	Description of how the baseline scenario is identified and description of the identified baseline scenario	PDD B.4			
6.12	Does the PDD identify the baseline, a scenario that represents the anthropogenic emissions by sources of GHG that would occur in the absence of the proposed CDM project activity?	/1/ /6/	In accordance with ACM0002, as the project activity is the installation of a new grid-connected renewable power plant/unit, the baseline scenario is the following: Electricity delivered to the grid by the project activity would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources, as reflected in the combined margin (CM) calculations described in the "Tool to calculate the emission factor for an electricity system" (version 2). This is correctly identified in the PDD, and this does represent the anthropogenic emissions by sources of GHG that would occur in the absence of the proposed CDM project activity.	OK	OK
6.13	Have the procedures/ steps to identify the most reasonable baseline scenario, as required by the methodology and applicable tools, been documented clearly in the PDD? Are all feasible and credible alternatives identified including but not limited to all the potential scenarios listed in the methodology?	/1/ /6/	The baseline scenario is defined in ACM0002. This is in accordance with the baseline scenario identified in the PDD. No further discussion of alternatives is required.	OK	OK
6.14	Are realistic different configurations or combinations of alternatives that may be able to provide similar outputs	/1/ /6/	The baseline scenario is defined in ACM0002. This is in accordance with the baseline scenario identified in the PDD. No	OK	OK

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	Checklist Question	Reference	Comment	Draft Conclusion OK/CAR/CL	Final Conclusion OK/ NOT OK
	and services considered?		further discussion of alternatives is required.		
6.15	Are all considered alternatives assessed for consistency with (enforced) mandatory laws and regulations?	/1/ /6/	The baseline scenario is defined in ACM0002. This is in accordance with the baseline scenario identified in the PDD. No further discussion of alternatives is required.	OK	OK
6.16	Does the list of alternatives include the project activity undertaken without being registered as a CDM project?	/1/ /6/	The baseline scenario is defined in ACM0002. This is in accordance with the baseline scenario identified in the PDD. No further discussion of alternatives is required.	OK	OK
6.17	If alternatives are excluded: - Is it shown that at least one credible and feasible alternative does not face a barrier? Is this reasonable? - If the remaining alternatives include the project undertaken without CDM, is sufficient justification provided to demonstrate the validity of the barrier? Is sufficient justification provided to demonstrate that the CDM alleviates the identified barriers that prevent the project? How has this been verified?	/1/ /6/	The baseline scenario is defined in ACM0002. This is in accordance with the baseline scenario identified in the PDD. No further discussion of alternatives is required.	OK	OK
6.18	If barriers are used to exclude baseline alternatives, can the barriers be considered real, i.e. is sufficient evidence/ justification provided to support every exclusion of alternatives? Is it reasonable?	/1/ /6/	Not applicable.	N/A	N/A
6.19	Can the barriers be considered preventative, i.e. do they really prevent the alternatives from taking place?	/1/ /6/	Not applicable.	N/A	N/A
6.20	If Investment Analysis is used to exclude baseline alternatives, has it been correctly applied? Are assumptions and input values reasonable and sufficiently justified?	/1/ /6/	The baseline scenario is defined in ACM0002. This is in accordance with the baseline scenario identified in the PDD. No further discussion of alternatives is required.	OK	OK
6.21	Is a transparent version of the analysis provided (i.e. financial analysis spreadsheets) and is in compliance with the 'guidelines on the assessment of investment analysis)?	/1/ /6/ /13/	Yes, the financial analysis spreadsheet is a transparent and it is in compliance with the 'guidelines on the assessment of investment analysis.	OK	OK
6.22	Have all relevant national and/or sectoral policies and circumstances been taken into account? Are they listed in the PDD?	/1/ /6/	Yes. The selected baseline scenario is in compliance with relevant national and sectoral policies. They are mentioned in the PDD.	OK	OK
6.23	Does the PDD provide a verifiable description of the baseline scenario, including a description of the technology/ activities that would have been employed in the absence of the CDM project?	/1/ /6/	Yes. The description is in accordance with the methodology ACM0002 version.	OK	OK

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	Checklist Question	Reference	Comment	Draft Conclusion OK/CAR/CL	Final Conclusion OK/ NOT OK
6.24	Does the identified baseline scenario reasonably represent what would occur in the absence of the proposed project activity?	/1/ /6/ /9/ /10/	Yes. In the absence of the project activity, electricity would continue to be generated by the existing capacity in the NWPG.	OK	OK
7.	Additionality (CDM VVM EB 55 (para.94-121))	PDD B.6			
	a) Prior consideration of the CDM	PDD C.1.1			
7.1	Is the start date defined in accordance with the "Glossary of CDM terms"? What evidence is provided to verify that this was the official start date? Is this considered reliable and reasonable?	/1//20//21//22//23/	CAR 2 In order to determine the starting date of project, the project participant would need to provide the construction contract to determine the starting date.	CAR-2	OK
7.2	Is it a new project activity (start date on or after August 2008) or an existing project?	/1//20//21//22//23/	The project is a new project activity, since the start date is later than 02 August 2008.	OK	OK
7.3	For a new project which does not require a new methodology and has not published its PDD for stakeholder comments prior to the start date, then: a. Have the project proponents informed the DNA and/or UNFCCC secretariat in writing? How has this notification been verified? (i.e. confirmation from the DNA or UNFCCC) b. Was the notification made within 6 months of the project activity start date? c. Does the letter/ notification indicate the precise geographic location and provide a brief description of the proposed project? d. Have the project proponents informed the DNA and/ or UNFCCC secretariat of the progress of the project activity every subsequent two years after the initial notification?	/1/ /22/ /23/	a. Yes. The project developer submitted a CDM prior consideration form to the UNFCCC Secretariat on 25 April 2010. A written notification to the China DNA was submitted on 25 March 2010 and approved on 6 April 2010. Both of the documents have been provided and verified by the validation team, and are considered to be reliable. b. Both notifications have been verified by ERM CVS. But the starting date of project is not defined clearly. It will be concluded after CAR 2 is closed. c. Yes. Both notifications state the precise geographic location, both state capacity, technology adopted and the grid to be connected. d. Not applicable.	CAR-2	OK
7.4	For an existing project which has a start date prior to the publication of the PDD for global stakeholder comments, has the project proponent provided the following: a. Evidence of awareness of the CDM prior to the project activity start date and that the benefits of the CDM were a decisive factor in the decision to proceed with the project? (e.g. Board minutes, notes etc) Is this sufficient? b. Reliable evidence that demonstrates real actions were taken to secure CDM status in parallel with the project's implementation? (e.g.		Not applicable	N/A	N/A

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	Checklist Question	Reference	Comment	Draft Conclusion OK/CAR/CL	Final Conclusion OK/ NOT OK
	contracts with consultants for CDM/PDD/methodology services, ERPAs, correspondence with CER buyers, DOEs, DNAs or the UNFCCC). Is this sufficient?				
	b) Identification of alternatives (Additionality Tool)	PDD B.5			
7.5	Is the assessment of alternatives in compliance with the requirements of the methodology and the relevant tool(s) (e.g. the Tool for the demonstration and assessment of additionality)? Is the assessment consistent with section B.4?	/1/ /12/	The PDD includes the identification of alternatives scenarios in section B.5, following step 1 of the "Tool for the demonstration and assessment of additionality". The identified alternatives to the project activity are (1) The project activity not undertaken as a CDM project; (2) Construction of a thermal power plant; (3) Construction of other renewable energy power plant; (4) Equivalent annual generated electricity supplied by NWPG. ERM CVS can confirm that alternatives are in compliance with the requirements of the methodology ACM 0002, and the Tool for the demonstration and assessment of additionality, version 5.2.	OK	OK
	c) Investment Analysis				
7.6	Has an investment analysis been used to demonstrate additionality? Is Investment analysis appropriate in this case to demonstrate the investment decision? (i.e. is financial attractiveness the key investment criteria?)	/1/ /13/	Yes, an investment analysis has been used to demonstrate additionality. Yes, the Investment analysis is appropriate in this case to demonstrate the investment decision of a private sector developer.	OK	OK
7.7	Has the project activity and investment decision been clearly defined/ framed? That is, has the CDM project activity been defined separately from the overarching project or facility and is the investment decision in this case clearly framed? (e.g. is the project to 'make cement' or is it to 'supply fuel to a cement factory'?)	/1/ /11/ /13/	Yes, the project activity and investment decision has been clearly defined/ framed.	OK	OK
7.8	Has the appropriate analysis Option been chosen? (as per the <i>Guidance on the Assessment of Investment Analysis</i>) - If Option I is chosen (simple cost analysis), is it demonstrated that the alternatives produce no economic benefits other than CDM income? - If Option II is chosen (investment comparison), does the proposed baseline scenario leave the PP no other choice than to make an investment to supply the same and (or substitute) products or services?	/1/ /13/	Since the proposed project generates income from the sale of electricity in addition to revenues from the sale of CERs, the simple cost analysis method (option I) is not appropriate. Since the alternative to the project (NWPG) as the provider for the same electricity output as the proposed project) does not involve investments of a comparable scale, option II is not applicable. Therefore the PDD has correctly identified option III (benchmark analysis).	OK	OK

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	Checklist Question	Reference	Comment	Draft Conclusion OK/CAR/CL	Final Conclusion OK/ NOT OK
	If Option III is chosen, is it appropriate in this case?				
7.9	Is the most suitable financial indicator clearly identified (Project or Equity IRR, NPV, cost benefit ratio, or (levelised) unit cost)?	/1//13//11/	Yes, the most suitable financial indicator, project IRR, is identified in accordance with the <i>Interim Rules on Economic Assessment of Electrical Engineering Retrofit Projects</i> , which is a widely used and publicly available reference issued by the State Power Corporation of China. The selected indicator is reasonable and adequately justified.	OK	OK
7.10	If Option I is chosen: Are the assumptions consistent for all alternatives assessed? If not, are the differences justified?		Not applicable	N/A	N/A
7.11	If Option II is chosen (investment comparison analysis): Are the assumptions for all alternatives compared consistent (including discount rates if applicable)?		Not applicable	N/A	N/A
7.12	If Option III is chosen: <i>Benchmark (BM) or Discount Rate (DR)</i> - If an IRR indicator is used, is the choice of BM type consistent with the type of IRR calculated? (e.g. a Project IRR benchmark is appropriated for a WACC or Project IRR analysis; an Equity IRR benchmark is appropriate for an Equity IRR analysis) - Is the BM or DR value justified with supporting evidence for its appropriateness? - Is an appropriate BM or DR value chosen that is relevant for the sector (i.e. electricity generation, cement manufacture, yeast manufacture, hydropower etc)? - Is an appropriate BM or DR value chosen that is relevant for the project activity (i.e. for this investor, country, risk of project, time of investment decision)? - Is the chosen benchmark conservative and in line with other BM or DRs used in current or previous projects by the same investor? (including the BM or DR used in Feasibility Studies or other financial analyses of the project activity)	/1//11//13/	- Yes. The project IRR benchmark matches the project IRR calculation. This has been verified against the IRR spreadsheet, FSR, and the <i>Interim Rules on Economic Assessment of Electrical Engineering Retrofit Projects</i>. - Yes. The benchmark is taken from the <i>Interim Rules on Economic Assessment of Electrical Engineering Retrofit Projects</i>, which is a widely used and publicly available reference issued by the State Power Corporation of China. - Yes, according to the <i>Interim Rules on Economic Assessment of Electrical Engineering Retrofit Projects</i>, the financial benchmark IRR of electrical industry in China is regulated as 8% of the total investment. Hence, 8% is chosen as the benchmark for the project activity, which is appropriate for a grid connected power sector project including wind power projects. - Yes, according to the <i>Interim Rules on Economic Assessment of Electrical Engineering Retrofit Projects</i> issued by the State Power Corporation of China, which was valid at the time of FSR writing and the time of investment decision, power sector projects including wind projects in China should apply this benchmark. The benchmark is also applied by the local government when deciding whether to approve feasibility study reports, and therefore it does form the basis for the decision whether or not to invest in the project. - Yes. According to the <i>Interim Rules on Economic Assessment</i>	OK	OK

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	Checklist Question	Reference	Comment	Draft Conclusion OK/CAR/CL	Final Conclusion OK/ NOT OK
			<i>of Electrical Engineering Retrofit Projects</i> issued by State Power Corporation of China, power sector projects in China should apply this benchmark. The benchmark is widely used by similar projects, including CDM projects, in the host country. ERM CVS has not identified any other publicly available comparable benchmarks that are lower, hence the chosen benchmark is considered to be conservative.		
7.13	Source of BM or DR If an external BM or DR has been used: a. Is the BM or DR based on publicly available data sources? Have these data sources been validated? b. Are the assumptions underlying the referenced BM or DR also applicable to this project?	/1//11//13/	a. The BM is based on publicly available data sources-Economical assessment methods and parameters for construction projects. b. The assumptions underlying the referenced BM are also applicable to this project	OK	OK
7.14	Source of BM or DR If an internal company BM or DR has been used: a. Is the project participant the only possible investor in the project? b. Is it sufficiently demonstrated that the internal benchmark has been used for similar projects with similar risk or would have been used for similar projects in the same sector and country/region? c. How has this been validated? d. Has a lower BM or DR been used in previous investment decisions by the project participant (in this project or similar others)? If so, are there verifiable circumstances that have led to a change in the BM or DR? e. Is the chosen BM or DR value appropriate in comparison with other comparable publicly available comparable BM and DRs? f. Is the BM or DR consistent with others used in similar projects (and validated by CVS)?		Not applicable	N/A	N/A
7.15	Risk Premiums a. Are risk premiums applied in the development of the BM or DR? b. If so, are they reasonable and justified? How has this been validated?	/1//11//13/	Not applicable. A government benchmark is used and no additional risk premiums are applied.	N/A	N/A
	Assumption and Input Values, calculations				

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	Checklist Question	Reference	Comment	Draft Conclusion OK/CAR/CL	Final Conclusion OK/ NOT OK
7.16	Are all references made in the investment analysis correctly referenced/ sourced? Have these sources been verified?	/1//4//19/	Yes, all references made in the investment analysis correctly referenced/ sourced. These reference sources have been cross-checked with FSR.	OK	OK
7.17	Have values from a feasibility study report (FSR) approved by national authorities been used? If so: a. Has the FSR been the basis of the decision to proceed with the investment in the project? How has this been verified? b. Are the values used in the PDD and associated annexes valid and consistent with the FSR? c. At the time of the investment decision, are the input values from the FSR valid and applicable (based on specific local and sectoral expertise and knowledge)?	/1/ /4/ /19/ /21/ /32/ /33/ /34/	a. Yes. The FSR is adopted as the basis of the decision to proceed with the investment in the project since it is in accordance with Chinese regulations; the FSR must be completed and approved before the project is permitted to be built. b. All values used in the PDD are consistent with the FSR. c. The total installed capacity of the proposed project is 201 MW. 5 previous windfarm CDM projects have already been registered with the same installed capacity in the same province (Reference numbers 3241, 3512, 4138, 4253 and 4278). Therefore ERM CVS has made a comparison of key input values against other registered wind projects in Gansu Province with the same installed capacity (201 MW). The data source is from the UNEP Risoe CDM pipeline /34/ and the PDDs and validation reports published on the website of the UNFCCC /33/ a. The tariff stated in the FSR is RMB 0.5206. According to the FSR approval issued by NDRC on 21 April 2009, the tariff for the proposed project is RMB 0.5206 (tax inclusive), indicating that the tariff remains valid and applicable at the time of investment decision. If the highest tariff 0.585 RMB/kWh (incl. VAT) in Gansu province, as applied by the Executive Board, is used in the IRR calculation sheet, the IRR is 7.64% which remains lower than the benchmark of 8%. b. The value is sourced from the third party approved FSR, and the unit cost of the proposed project is calculated as 9,493 RMB/kW. ERM CVS has confirmed the information against the FSR /4/. According to the UNEP Risoe CDM pipeline /34/ and the website of UNFCCC /33/, the unit cost of other wind power projects with the same installed capacity in the same region (Gansu province) varies from 9,461 RMB/kW to 9,851 RMB/kW. And the average level of unit cost is 9,571 RMB/kW. The total investment cost of the proposed project is within the range of similar projects in the region, and it is slightly lower than the average level. In order to further cross check and confirm the investment costs, the principal investment related third party contracts were reviewed /19//20/. The total contracted value is approximately 1,840	CL	OK

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	Checklist Question	Reference	Comment	Draft Conclusion OK/CAR/CL	Final Conclusion OK/ NOT OK
			million RMB. ERM CVS was able to review the cost breakdown of the same components in the FSR and the overall variance between the contracts and the FSR estimation of the cost is 3.56%, which shows that the estimation in the FSR is reasonable given that the project was still under construction and further expenditure was still yet to be incurred. Therefore the estimated value in the FSR is considered reasonable. c. The annual average O&M costs of the project were compared against other similar registered projects of the same installed capacity in the region. These projects were identified according to UNEP Risoe CDM pipeline /34/ and website of UNFCCC /33/. The Unit O&M costs (RMB/MWh) of this project are 115 RMB/MWh on average, which are slightly higher than the average of 92.9 RMB/MWh for previous projects. This represents a difference of 23.7% above the average level of the 5 previously registered projects. However as discussed in the sensitivity analysis section, annual O&M cost would need to decrease by more than 62.5% in order for the IRR to exceed the benchmark. ERM CVS has reviewed the China Statistical Yearbook /21/, and can confirm that this is very unlikely to happen considering the labour and material costs have shown a steadily increasing trend with inflation and China's economic growth in recent decades. So ERM CVS confirm that the annual average O&M costs is reasonable. d. The annual power delivered to the grid is 461,464 MWh which is calculated by the third party design institute in the FSR and approved by the National government /4/. The meteorological data was monitored by Guazhou station with year from 1978-2007 /4/. ERM confirm that the annual power is reasonable with long time-series monitoring. According to the UNEP Risoe CDM pipeline /34/ and the website of the UNFCCC /33/, the annual power delivered to the grid is higher than the average of previously registered projects with the same installed capacity in the same province, which is 445,692 MWh. e. Load factor: Refer to CL 1		
7.18	<i>Technical assumptions</i> a. Are the technical assumptions reasonable?	/1//4//19/	a. CL 1: The following technical information is not provided in section A.4.3 of the PDD, as required by the PDD guideline	CL	OK

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	Checklist Question	Reference	Comment	Draft Conclusion OK/CAR/CL	Final Conclusion OK/ NOT OK
	b. Are the assumptions adequately supported by evidence/ justification? c. What evidence has been provided to support critical technical assumptions? Have technical assumptions and input values been verified by: assessing them against the available evidence and expertise; cross-checking the parameters against 3rd party or publicly available sources; reviewing feasibility reports; reviewing information of other similar projects; reviewing project information presented in permit applications etc; referring to a sector or technical expert; etc?		(1)the lifetime of the project equipment and the reference for the project operational lifetime should be added in section C.1.2; (2)monitoring equipments; (3)technology transfer; (4)load factor; and (5)GHGs emission sources. (6) effective electricity coefficient b. This will be concluded after CL 1 is closed. c. This will be concluded after CL 1 is closed [Note: points (b) and (c) above have been concluded – the critical technical assumptions can be confirmed and have been validated in section 3.3, 3.6 and CL 1 – please refer to the relevant sections of the report]		
7.19	<i>Financial Assumptions</i> a. Are the revenue and price financial assumptions reasonable? b. Are the assumed costs reasonable? Do they all accrue to the CDM project activity only (rather than to other parts of the facility)? c. Are all the assumed taxes applicable for the different alternatives and applicable for the whole assessment period? d. Are the assumptions adequately supported by evidence/ justification? What evidence has been provided to support critical financial assumptions? Have the financial assumptions and input values been verified by checking them against feasibility studies, quotes, receipts, third-party forecasts, annual reports, and financial analyses (such as those presented to banks), etc?	/1/ /4/ /17/ /28/ /30/ /32/ /33/ /34/	Techno-economic parameters and assumptions, such as investment costs, operational cost, and project lifetime have been checked and ERM CVS can confirm the following: a. Revenue generated by the project is based on the annual power output and electricity tariff. The electricity tariff applied in the FSR is the approved tariff by National Development and Reform Commission (NDRC) on 21 April 2009. Annual power output is determined by the independent third party design institute (Northwest Hydro Consulting Engineers) based on a long time-series (30 years) of meteorological data, and one year of on site measurement. But the load factor should be addressed in the PDD b. Estimated investment cost is derived from the FSR, and it is assessed as reasonable by cross checking against the other CDM projects in the same region according to the CDM pipeline published by UNFCCC and UNEP Risoe CDM pipeline. Investment costs are found to be close to the average of other similar projects. CL 4 The validation team has verified the investment cost against the approved FSR and deemed to be the same, however, please provide further evidence to cross check the investment cost c. Taxes applicable to the project are: corporate income tax, VAT (Value added tax), and surplus taxes. All tax rates	CL1 CL4	OK

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	Checklist Question	Reference	Comment	Draft Conclusion OK/CAR/CL	Final Conclusion OK/ NOT OK
			applied by the project are in line with executive laws/regulations in China at the time of investment decision and are applicable d. Financial input parameters are all derived from the approved third party FSR. Further details of cross checks can be found in section 3.5 of the validation report.		
7.20	<i>Timing of assumptions</i> a. Are all assumed input values valid for the time of the investment decision? b. Are all capex costs valid at the time of the investment decision? Are there any sunk costs? c. Are all revenues and costs reasonable for the whole period as forecast? d. Are changes in costs or revenues scheduled or likely? Have such changes been incorporated and justified? e. Are the costs and revenues entered in the correct year when they will occur?	/1/ /4/ /17/ /28/ /30/ /35/	a. According to the VVM paragraph 113, the FSR has been the basis of the decision to proceed with the investment in the project. The FSR was prepared in January 2009 and approved in April 2009. The validation team has verified the investment cost against the approved FSR and deemed to be the same, however, please provide further evidence to cross check the investment cost which refers to CL 4. b. All capex costs are valid at the time of the investment decision. No sunk costs are relevant for the project. c. See above comment in bullet point (a) of this column d. The proposed project is under construction. No changes in key input parameters have been identified or are considered likely. e. Yes, costs and revenues are accounted for according to the timetable for project development as set out in the approved third party design institute FSR.	CL4	OK
7.21	<i>Revenues</i> Are all benefits of all the assessed alternatives incorporated in the analysis? (e.g. including revenues from by-products, reduced costs etc; consult with sector expert)	/1//4/	The benefits of all the assessed alternatives are incorporated in the analysis. The only revenue stream for the project is the sales of electricity to the grid.	OK	OK
7.22	<i>Costs</i> Are all costs of all the assessed alternatives incorporated in the analysis? (e.g. including permit and licence costs, transport costs etc; consult with sector expert)	/1//4/ /21/ /33/ /34/	The validation team has verified the investment cost against the approved FSR and deemed to be the same, however, please provide further evidence to cross check the investment cost. The annual average O&M costs of the project were compared against other similar registered projects of the same installed capacity in the region. These projects were identified according to UNEP Risoe CDM pipeline /34/ and website of UNFCCC /33/. The Unit O&M costs (average annual O&M costs in RMB per MWh electricity delivered to the grid) of this project are 115 RMB/MWh on average, which are slightly higher than the	CL4	OK

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			average of 92.9 RMB/MWh for previous projects. This represents a difference of 23.7% above the average level of the 5 previously registered projects. However as discussed in the sensitivity analysis section, annual O&M cost would need to decrease by more than 62.5% in order for the IRR to exceed the benchmark. ERM CVS has reviewed the China Statistical Yearbook /21/, and can confirm that this is very unlikely to happen considering the labour and material costs have shown a steadily increasing trend with inflation and China's economic growth in recent decades.		
7.23	Are there any policies, subsidies, incentives, grants, tax breaks etc that apply to any of the alternatives? Are these incorporated in the analysis? (refer to <i>Clarifications on the consideration of national and /or sectoral policies and circumstances in baseline scenarios</i> , currently located at http://cdm.unfccc.int/EB/022/eb22_repan3.pdf)		The IRR calculation of the proposed project is in compliance with all national and sectoral policies. These are no additional incentives such as tax breaks that need to be included in the analysis.	OK	OK
7.24	Is the assumed period of assessment appropriate? How has this been demonstrated? (i.e. based on economic lifetime of equipment/ assets or other dependent factors)	/1//4//19/	Yes. The expected lifetime of the equipments is 22 years, based on the equipment purchase contract.	OK	OK
7.25	Is any residual value of the project activity assets included in the analysis? Are residual value assumptions reasonable and justified and consistent with local accounting rules, international best practice and industry experience?	/1//3//4/	Yes. Residual value of fixed assets is included in the analysis. The analysis is carried out over a period of 22 years, and 5% residual value is included at the end of the period. This is consistent with the Chinese accounting rules, and has been verified in the FSR. 5% residual value is also in line with the common practice for other projects in China based on ERM CVS's local, sectoral and financial expertise.	OK	OK
	Calculations				
7.26	Has the project participant supplied unprotected and traceable spreadsheet versions of all investment analysis?	/1//3/	Yes, the project participant has supplied unprotected and traceable spreadsheet versions of all investment analysis.	OK	OK
7.27	From the investment analysis provided, is it possible to reproduce the results?	/1//3/	Yes, it is possible to reproduce the results.	OK	OK
7.28	Have the listed input values been consistently applied in all calculations?	/1//3/	Yes, the listed input values have been consistently applied in all calculations.	OK	OK
7.29	Are the computations/ formula correct? (this includes the computations implicit in input values, such as technical calculations of the amount of energy demanded or sold etc)	/1//3/	Yes, the computations/ formula are correct.	OK	OK

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7.30	<i>Depreciation</i> a. Are deprecation costs applied to depreciable assets only (not land)? b. Are the depreciation and major repair and maintenance costs consistent with the assessment period and the residual values? c. Are depreciation costs/ periods consistent with local accounting regulations? d. Are depreciation costs (and other non-cash items) related to the project activity <u>excluded (not deducted)</u> from net Cash Flow used for calculating the financial indicator (e.g. IRR, NPV)?	/1/ /2/	Depreciation is correctly accounted for. Depreciation and interest payments are only used for the calculation of income tax, and are not included as cash expenditures in the final cash flow. This is conservative, since the inclusion of these parameters in the calculation of income tax leads to a lower tax liability, and consequently a higher IRR. a. Yes. b. The depreciation calculations are consistent with the assessment period and residual value and are in line with the FSR. However: Minor Issue 1: Spreadsheet refers to depreciation as "devalue". The correct term shall be used. Equipment maintenance costs are being considered as 1.5% over total investment (years 3-5); 2.0% (years 6-8); and 2.5% (years 9-25). These costs do not follow the decline in electricity generation for the last two years. CL 5: The inclusion of 100% of maintenance costs for the last two years shall be explained, bearing in mind that electricity generation declines in these years. c. Yes, it is consistent. d. Depreciation was not included in the free cash flow assessment as an expense, it was only considered for tax calculation purposes.	CL 5 Minor Issue 1	OK
7.31	<i>Tax</i> a. Is the treatment of taxation consistent with the chosen benchmark or discount rate? (i.e. taxation should only be treated as an expense in the IRR/NPV calculation if the chosen BM or DR is intended for post-tax calculations) For post-tax BMs or DRs: b. Are interest costs included in the calculation of net taxable income and thus tax? c. Are interest costs calculated in accordance with	/1/ /2/ /11/ /17//28//29// 30/	The taxation adapted for the project activities is consistent with the local Chinese regulation, which is consistent with the chosen benchmark. Taxation is only included as an expense in the IRR calculation since the benchmark is intended for post-tax comparisons. As the Interim Rules on Economic Assessment of Electric Engineering Retrofit Projects specify that the benchmark is after taxes, it is comparable with the project IRR. Corporate tax rate (income tax) is 25% and it is derived from <i>Enterprise Income Tax Law of the People's Republic of China</i> (promulgated on Document Order of the President of the	OK	OK

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	Checklist Question	Reference	Comment	Draft Conclusion OK/CAR/CL	Final Conclusion OK/ NOT OK
	the <i>Guidance on the Assessment of Investment Analysis</i> d. Are depreciation costs included in the calculation of net taxable income and thus tax?		People's Republic of China (No. 63) on March 16, 2007) which is effective since 1 Jan 2008. The FSR of the proposed project was finalized in January 2009, so the tax rate is applicable.		
7.32	Interest costs If a Project IRR has been used, are the costs of financing expenditures (i.e. loan repayments and interest) excluded from the calculation of Project IRR? (financing costs should not be deducted from Net Cash Flow) If an Equity IRR has been used, is the debt portion of the investment cost excluded as a cash outflow and the interest costs and principal repayments included as costs?	/1/ /2//13//28//2 9//30/	Interest costs only apply for tax calculation purposes. Loan repayments were not included in the free cash flow calculation. Construction loan interests are not traceable and explanation on its calculation shall be provided. CL 6: Explanation on the construction loan interests shall be provided. The benchmark is a project IRR and calculations are in accordance with the chosen benchmark.	CL-6	OK
7.33	Recommended project: If the implementation of the project ceased and then recommenced due to consideration of the CDM, then: a. Are input values valid and applicable at the time of making the decision to recommence the project? b. Are capital costs incurred prior to the revised project activity start date input as the recoverable value of the assets (limited to the potential reuse/ resale of tangible assets)? c. How has the fair market value of the capital expenditures been calculated and validated? (e.g. by chartered specialists). Is this fair market value reasonable and justified?		Not applicable	N/A	N/A
7.34	Sensitivity analysis: a. Are all variable and critical costs and revenues in the analysis included in the sensitivity analysis? b. Is the assessed range of variations reasonable in light of the reliability of the estimated input values and the likely range? c. If some variations create scenarios that change the conclusion/ result of the analysis, how likely/ probable are such scenarios (in the opinion of the DOE)?	/1/ /2//13/	a. Yes. The sensitivity analysis was shown in the PDD for the most significant input values: Static Investment, O&M Costs, Tariff and Annual Output. b. The sensitivity analysis considers a range of +/- 10% for static investment, annual O&M cost, on-grid tariff and annual generation. The results show that without CER revenues the IRR for the project will not reach the benchmark under +/-10% variation. CL 7: Please provide the results of sensitivity analysis on what variations are needed for static investment, O&M	CL-7	OK

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	d. Is the sensitivity analysis possible to reproduce?		costs, tariff and annual output to change, in order to reach the benchmark for project IRR in the PDD. c. Except for variations on the tariff, the range considered in the remaining scenarios seems to be acceptable. This will be further assessed upon the resolution of CL 7. d. The sensitivity analysis can be reproduced and it is in accordance with the values in the PDD.		
	d) Barrier Analysis (VVM EB 55 para. 115 – 118)	PDD Step 3			
7.35	Has a barrier analysis been used?	/1/	No. A barrier analysis was not adopted and this is in line with the tool for the demonstration and assessment of additionality.	N/A	N/A
7.36	Is a complete list of credible, feasible and legally compliant alternatives identified?		Not applicable	N/A	N/A
7.37	Is a complete list of barriers that prevent the alternatives and the proposed CDM project activity from occurring identified?		Not applicable	N/A	N/A
7.38	Do any such identified barriers have a clear and direct impact on the financial returns of the project activity? (these are not barriers and should be assessed in the investment analysis)		Not applicable	N/A	N/A
7.39	Are the identified barriers real and substantiated by independent sources of data such as relevant national legislation, surveys of local conditions and national or international statistics? How has the reliability and credibility of the sources and assumptions used been assessed?		Not applicable	N/A	N/A
7.40	For each barrier, is at least one of the following types of evidence provided that is relevant and substantiates the identified barrier? (a) Relevant legislation, regulatory information or industry norms; (b) Relevant (sectoral) studies or surveys (e.g. market surveys, technology studies, etc) undertaken by universities, research institutions, industry associations, companies, bilateral/multilateral institutions, etc; (c) Relevant statistical data from national or international statistics;		Not applicable	N/A	N/A

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	Checklist Question	Reference	Comment	Draft Conclusion OK/CAR/CL	Final Conclusion OK/ NOT OK
	(d) Documentation of relevant market data (e.g. market prices, tariffs, rules); (e) Written documentation of independent expert judgments from industry, educational institutions (e.g. universities, technical schools, training centres), industry associations and others.				
7.41	Is the proposed project taking place in a least developed country (LDC)? In this case it is sufficient to transparently describe the relevant barriers, as less stringency is needed with regards to data availability in the actual demonstration of barriers. Projects in LDCs are not bound by the provisions in the 'Guidelines for objective demonstration and assessment of barriers' (questions 7.42 to 7.45) and may use other approaches that are more adapted to the local circumstances.		Not applicable	N/A	N/A
7.42	If demonstrating barriers related to the lack of access to capital, technologies and skilled labour, do the project proponents provide information on the nature of the companies and entities involved in the financing and implementation of the project in accordance with guideline 4 of the 'Guidelines for objective demonstration and assessment of barriers'? How has this information been validated? Does the information on the nature of the companies/entities involved lend credibility to the claimed barriers relating to lack of access to capital, technologies or skilled labour?		Not applicable	N/A	N/A
7.43	Has the PP demonstrated, for each of the barriers, that in similar circumstances (in similar industries/sectors, in companies of similar size and ownership structure, in similar projects) the barriers actually prevented the implementation of other project(s)? Note that this approach is not mandatory and that other approaches to enhance objectivity of barrier analysis may also be pursued. (Guideline 3 of the 'Guidelines for objective demonstration and assessment of barriers')		Not applicable	N/A	N/A
7.44	If barriers related to increased risks of damage (i.e. that the equipment is damaged due to technological barriers, lack of know-how etc.) are claimed, have these been		Not applicable	N/A	N/A

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	Checklist Question	Reference	Comment	Draft Conclusion OK/CAR/CL	Final Conclusion OK/ NOT OK
	quantified by the calculation of probability of loss and loss expenses, and can the underlying data and assumptions can be objectively and transparently justified? Note: This quantitative approach to barriers is an option in case sufficient data is available, as a limited number of projects may have the data to follow this approach (Guideline 5 of the 'Guidelines for objective demonstration and assessment of barriers').				
7.45	If PPs claim investment barriers, does the PDD demonstrate that the financing of the project was assured only due to the benefits of the CDM? Is it demonstrated that the loan approval (or other significant financing decision(s)) by the lender takes explicitly the CDM registration into account? (Guideline 6 of the 'Guidelines for objective demonstration and assessment of barriers').		Not applicable	N/A	N/A
7.46	Based on the evidence reviewed, and conservative interpretations of this evidence, can it be confirmed that the identified barriers are real?		Not applicable	N/A	N/A
7.47	Based on the evidence reviewed, and conservative interpretations of this evidence, can it be confirmed that the identified barriers actually prevent the implementation of the proposed CDM project activity by the project participant or other potential project participants? How has this been validated?		Not applicable	N/A	N/A
7.48	Is at least one of the alternatives remaining (i.e. is not prevented by the barriers)? How has this been validated?		Not applicable	N/A	N/A
7.49	Is it clearly explained how approval of the project in the CDM would enable the proposed project activity to surmount the barrier, in an objective way? Is the rationale reasonable and justified with transparent and documented evidence, and conservative interpretations of this evidence?		Not applicable	N/A	N/A
7.50	Overall, is the Barrier Analysis presented credible and compliant with the applicable Tools?		Not applicable	N/A	N/A
	e) Common Practice Analysis (VVM EB 55 para. 119 – 121)	<i>PDD Step 4</i>			
7.51	Is the proposed project activity a 'first of its kind'? Has	/1/	No, the PDD does not claim that the project is the first of its kind.	OK	OK

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	sufficient evidence been provided to validate this claim, and how have we validated this evidence?				
7.52	Has common practice analysis been undertaken and fully described in the PDD?	/1/	Yes	OK	OK
7.53	Is the geographical scope of the common practice analysis appropriate for the assessment related to the project activity's technology or industry type? If a region other than the host country is chosen, is this appropriate?	/1/	Yes, the geographical scope of the common practice analysis is selected as Gansu Province. This is considered reasonable since each province / autonomous region in China is large in terms of area and population size, and has authorities for its own policy framework, and tariffs and regulatory environments vary significantly between provinces / autonomous regions in China.	OK	OK
7.54	Has the scope of the comparison been defined correctly, i.e. how are 'similar' projects defined and is the definition of 'similar' appropriate, i.e. the same country/region, broadly similar technology, similar scale, take place in a comparable environment with respect to regulatory framework, investment climate, access to technology, access to financing, etc.	/1/	Yes, the geographical scope of the common practice analysis is selected as Gansu Province. This is considered reasonable since each province / autonomous region in China has authorities for its own policy framework, and tariffs and regulatory environments vary significantly between provinces / autonomous regions in China. Projects with capacity varying from 100MW to 300MW (a range of +/- 50%) have been considered as similar projects, which is considered to be reasonable.	OK	OK
7.55	Have all comparable projects been included in the common practice analysis? Has the PP provided Provide documented evidence and, where relevant, quantitative information? How was this assessed (by the DOE)?	/1/ /32/	The 2010 China Wind Power Development Report and the web site of UNFCCC listing those projects that applied for CDM in Gansu Province have been checked. Yes. The exclusion of non-comparable projects is reasonable.	OK	OK
7.56	If non availability of data has been used as a reason to exclude consideration of similar projects, how has this been validated?		N/A	N/A	N/A
7.57	Have similar and operational projects other than CDM project activities been undertaken in the region?	/1/	Yes, Gansu Anxi (Jiuquan phase 1) Wind Farm Project, 100.5 MW exists, however this project has been registered as Gold Standard project, which is verified from the following weblink: https://gs1.apx.com/mymodule/ProjectDoc/EditProjectDoc.asp?id=554	OK	OK
7.58	Are these widely observed and commonly carried out? If so: a. How have the essential distinctions with the proposed CDM project activity been assessed, e.g. that explain why the similar activities enjoyed certain benefits that rendered it financially /economically	/1/	No. Only one similar project can be identified in the geographical scope. a. The project is non-CDM, but, due to the low tariff obtained by the project, they applied for Gold Standard (GS) approval and the project was successfully registered under the Gold Standard.	OK	OK

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	attractive (e.g., subsidies or other financial flows) and which the proposed project activity cannot use or did not face the barriers faced by the proposed project? Are such distinctions considered fundamental? b. Are such distinctions verifiable, i.e. justified with sufficient evidence? c. If inaccessibility of data is the reason why some projects have not been included in the analysis, is justification of this claim provided?		b. It was confirmed by the validation team through assessment of the GS registry. c. Not applicable.		
7.59	Overall, is the proposed CDM project activity considered common practice?	/1/	No. It is the validation team's opinion that the proposed project activity cannot be considered common practice in the geographical region.	OK	OK
8.	Emissions Reductions (CDM VVM EB 55 para. 89 - 93)	PDD B.6			
	<i>Explanation of methodological choices</i>				
8.1	Is it explained how the procedures provided in the Methodology and applicable Tools are applied by the proposed project activity? (i.e. Are the required steps clearly followed?)	/1/ /6/ /9/	Yes. This has been verified by cross-checking the PDD against the methodology and the <i>Tool to Calculate Emission Factor for an Electricity System</i> version 02, in accordance with ACM0002 version 12.1.0.	OK	OK
	<i>Project emissions:</i>				
8.2	Is every choice of options for calculating project emissions offered by the methodology correctly justified? Is this justification in line with the situation as evidenced by site visits, local knowledge and supporting documentation?	/1/ /6/	Yes. Project emissions are zero as the proposed project is a new-grid connected power plant utilizing renewable energy according to the methodology ACM0002.	OK	OK
8.3	Are the formulae required for the determination of project emissions correctly presented, enabling a complete identification of parameters to be used and / or monitored?	/1/ /6/	NA	N/A	N/A
	<i>Baseline emissions:</i>				
8.4	Is every choice of options for calculating baseline emissions offered by the methodology correctly justified? Is this justification in line with the baseline scenario?	/1/ /6/ /9/	Yes. This has been verified by cross-checking the PDD and the <i>Tool to Calculate Emission Factor for an Electricity System</i> version 2, in accordance with ACM0002.	OK	OK
8.5	Are the formulae required for the determination of baseline emissions correctly presented, enabling a	/1/ /6/ /9/ /10/ /14/	<u>Operating Margin (OM)</u> : The Simple OM is used. This is appropriate since low-cost/must run resources constitute less	OK	OK

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	complete identification of parameters to be used and / or monitored?	/15/ /16/	than 50% of total grid generation in the average of the five most recent years (2003-2007). The Simple OM emission factor is calculated as the generation-weighted average emissions per electricity unit (tCO₂/MWh) of all generating sources serving the system, excluding low operating cost and must-run power plants. The data on fuel consumption and net electricity generation of each power plant /unit in the grid is not publicly available, therefore the simple OM is calculated based on data on the total net electricity generation of all power plants serving the system and the fuel types and total fuel consumption of the project electricity system (option B). Aggregated generation and fuel consumption data are used as more disaggregated data are not available. Low-cost/must run power resources in the grid include only renewable power generation, and the quantity of electricity supplied to the grid by these sources is known. Off-grid power plants are not included in the calculation. Therefore, Option B is appropriate for calculating the Simple OM emission factor. Net calorific values of each fuel type were obtained from the China Energy Statistical Yearbook, and IPCC 2006 default values were used for the oxidation factor and emission factor of each type of fossil fuel. The values used and the calculation of the simple OM is considered to be reasonable, and is in line with official data published by the Government of China.. <u>Build Margin (BM)</u>: Because plant specific fuel consumption and electricity generation data are not publicly available in China, the guidance given by the CDM Executive Board for a deviation from methodology AM0005 has been applied for calculation of the build margin (BM) emission factor for this project. In accordance with this guidance, the build margin consists of the set of power capacity additions in the electricity system that comprises 20% of the generation capacity (in MW) of the system, that have been built most recently, based on the aggregate incrementally installed capacity of all generation sources in year y, and the aggregate incrementally installed capacity of all generation sources in year y-n, where n represents the number of years of historical data that need to be considered in order for the sample group to comprise 20% of the total system generation capacity (in MW). The emissions factor of fossil fuel fired power generation in the grid is calculated using the proportions of GHG emissions from solid, liquid and gaseous fuels in the total GHG emissions related to power generation as the weights, and the emission factors of the most advanced commercial generation		

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			technologies available in the host country (as published by the NDRC). Finally, the BM emission factor is calculated as the product of this emission factor of fossil fuel fired power generation and the proportion of fossil fuel fired power plants in the newly installed 20% capacity, based on data for years 2005-2007, contained in the power yearbooks 2006-2008. The OM and BM of the connected regional grid system (NWPG) are calculated based on public and official data from the China Energy Statistical Yearbooks 2006-2008 editions and the China Electric Power Yearbooks 2006-2008 editions. The weights ω_{OM} and ω_{BM} are selected as 0.75 and 0.25 respectively for wind projects according to the <i>Tool to calculate the emission factor for an electricity system</i>. The combined margin is fixed ex-ante for the entire first crediting period. The results are consistent with the data published by the DNA of China in the China Regional Grid Baseline Emission Factor published in 2009.		
8.6	Are the applicable Tools and methods to calculate parameters correctly applied?	/1/ /9/	Yes, the baseline grid emissions factor is correctly calculated in line with the tool, and the validated grid calculations published by the DNA of China. The calculations adopt the accepted deviation approach to AM0005, which also applies to ACM0002.	OK	OK
8.7	Are the applicable parameters and equations correctly applied?	/1/ /9/	Yes	OK	OK
	<i>Leakage:</i>				
8.8	Are all potential sources of leakage correctly identified for the applied Methodology?	/1/ /6/ /9/	Not applicable. According to the methodology ACM0002, leakage does not need to be considered.	N/A	N/A
8.9	Are the formulae required for the determination of leakage emissions correctly presented, enabling a complete identification of parameters to be used and / or monitored?		Not applicable	N/A	N/A
8.10	Are the applicable Tools and methods for calculating leakage correctly applied?		Not applicable	N/A	N/A
8.11	Are the applicable parameters and equations correctly applied?		Not applicable	N/A	N/A
	<i>Emissions Reductions:</i>				
8.12	Are the parameters and equations used to calculate emission reductions applicable? Are the applicable parameters and equations correctly	/1/ /6/ /9/	Yes. The parameters and equations used to calculate emission reductions are applicable and in accordance with the <i>Tool to Calculate the Emission Factor for an Electricity System version 2</i>	OK	OK

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	applied?		and the methodology ACM0002. The equations and parameters are correctly applied.		
	Data and Parameters	PDD B.6.2			
8.13	Is the list of parameters presented in chapter B.6.2 of the PDD considered to be complete with regard to the requirements of the applied methodology?	/1/ /6/ /9/ /10/	Yes. This has been verified against the <i>Tool to Calculate Emission Factor for an Electricity System</i> version 2	OK	OK
8.14	What evidence is available to validate the accuracy and appropriateness of assumptions, data and parameters used in the calculation of project emissions ? Are the values used considered reasonable in the context of the proposed CDM project activity?	/1/ /6/ /9/ /10/ /15/	The data used in the baseline emission calculation is based on the data presented in the <i>China Electric Yearbook 2006-2008</i> . All parameters and equations stated in the PDD have been verified by cross-checking data and equations published by the China NDRC stated in the <i>2009 Baseline Emission Factors for Regional Power Grid in China</i> and <i>Appendix 3: Baseline Information</i> .	OK	OK
8.15	For each parameter: i. Title in line with Methodology? j. Data unit correctly expressed? k. Appropriate description? l. Source clearly referenced? (appropriate?) m. Correct value provided? n. Has this value been verified? o. Choice of data correctly justified? p. Measurement method correctly described?	/1/ /6/ /9/ /10/ /14/ /15/ /16/	$FC_{i,y}$ a. Yes. The title is in line with the methodology ACM0002 and Tool to calculate the emission factor for an electricity system v 2. b. Yes. The data unit is correctly expressed. c. Yes. Appropriate description is stated, which is "The amount of fuel i (in a mass or volume unit) consumed by relevant power sources m in year(s) y." d. Yes. All sources are clearly referenced. e. Yes. Correct values are provided in the Annex 3 of the PDD. f. Yes. All values have been verified against the China Energy Statistical Yearbook (2006-2008). g. Yes, the latest data available at the time of validation submission, 2006-2008, were used for the calculation, which is the recent 3 years data available at that time. h. not applicable. $NCV_{i,y}$ a. Yes. The title is in line with the methodology ACM0002 and Tool to calculate the emission factor for an electricity system v 2. b. Yes. The data unit is correctly expressed. c. Yes. Appropriate descriptions are stated, which is net calorific value (energy content) of fossil fuel type i. d. Yes. All sources are clearly referenced. e. Yes. Correct values are provided in the Annex 3 of the PDD.	OK	OK

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			f. Yes. All values have been verified against the China Energy Statistical Yearbook (2008). g. Yes, the latest data available at the time of validation submission used for the calculation. h. not applicable <i>EF_{CO2,i,y}</i> a. Yes. The title is in line with the methodology ACM0002 and Tool to calculate the emission factor for an electricity system v 2. b. Yes. The data unit is correctly expressed. c. Yes. Appropriate description is stated, which is CO₂ Emission Factor of fossil fuel type <i>i</i> in year <i>y</i> d. Yes. All sources are clearly referenced. e. Yes. Correct values are provided in the Annex 3 of the PDD. f. Yes. All values have been verified against 2006 IPCC Guidelines for National Greenhouse Gas Inventories g. Yes, Values from 2006 IPCC is the latest available values. h. not applicable <i>EG_y</i> a. Yes. The title is in line with the methodology ACM0002 and Tool to calculate the emission factor for an electricity system v 2. b. Yes. The data unit is correctly expressed. c. Yes. Appropriate description is stated, which is "The power supplied to the grid from Shaanxi, Gansu, Qinghai, Ningxia and Xinjiang in 2003-2007." d. Yes. All sources are clearly referenced. e. Yes. Correct values are provided in the Annex 3 of the PDD. f. Yes. All values have been verified against the China Electric Power Yearbook (2006-2008). g. Yes, the latest data available at the time of validation submission, 2006-2008, were used for the calculation, which are the recent 3 years of data available at that time. h. not applicable <i>Auxiliary Power Ratio</i>		

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	Checklist Question	Reference	Comment	Draft Conclusion OK/CAR/CL	Final Conclusion OK/ NOT OK
			a. Yes. The parameter is used for the calculation of the emissions factor in China according to the accepted deviation to AM0005 which also applies to ACM0002 b. Yes. The data unit is correctly expressed. c. Yes. Appropriate description is stated, which is “The auxiliary power ratio of source j in Shaanxi, Gansu, Qinghai, Ningxia and Xinjiang in 2005-2007” d. Yes. All sources are clearly referenced. e. Yes. Correct values are provided in the Annex 3 of the PDD. f. Yes. Value has been verified against the source from China DNA. g. Yes, the latest data available at the time of validation submission was used. h. Reference to measurement procedure is correct. <i>EF_{coal,adv}</i> a. The parameter is used for the calculation of the emissions factor in China according to the accepted deviation to AM0005 which also applies to ACM0002. b. Correctly expressed with provided value, not mentioned, add percentage % as value. c. Yes. Appropriate description is stated, which is “Best electricity supply efficiency for coal fired plant.” d. Yes. All sources are clearly referenced. e. Yes. Correct value is provided. f. Yes. Value has been verified against the source from China DNA. g. Yes, the latest data available at the time of validation submission was used. h. Reference to measurement procedure is correct. <i>EF_{oil,adv}</i> a. The parameter is used for the calculation of the emissions factor in China according to the accepted deviation to AM0005 which also applies to ACM0002..		

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	Checklist Question	Reference	Comment	Draft Conclusion OK/CAR/CL	Final Conclusion OK/ NOT OK
			b. Correctly expressed with provided value, not mentioned, add percentage % as value. c. Yes. Appropriate description is stated, which is “efficiency level of the best technology commercially available in China for gas-fired and oil-fired power generators.” d. Yes. All sources are clearly referenced. e. Yes. Correct value is provided. f. Yes. Value has been verified against the source from China DNA. g. Yes, the latest data available at the time of validation submission was used. h. Reference to measurement procedure is correct. <i>EF_{gas,adv}</i> a. The parameter is used for the calculation of the emissions factor in China according to the accepted deviation to AM0005 which also applies to ACM0002. b. Correctly expressed with provided value, not mentioned, add percentage % as value. c. Yes. Appropriate description is stated, which is “Best electricity supply efficiency for coal fired plant.” d. Yes. All sources are clearly referenced. e. Yes. Correct value is provided. f. Yes. Value has been verified against the source from China DNA. g. Yes, the latest data available at the time of validation submission was used. h. Reference to measurement procedure is correct. <i>CAP_{i,y}</i> a. The parameter is used for the calculation of the emissions factor in China according to the accepted deviation to AM0005 which also applies to ACM0002. b. Yes. The data unit is correctly expressed. c. Yes. Appropriate description is stated, which is “Installed capacity in each province of NWPG.”		

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	Checklist Question	Reference	Comment	Draft Conclusion OK/CAR/CL	Final Conclusion OK/ NOT OK
			d. Yes. All sources are clearly referenced. e. Yes. Correct values are provided in the Annex 3 of the PDD. f. Yes. Value has been verified against the China Electric Power Yearbook (2006~2008). g. Yes, the latest data available at the time of validation submission was used. h. Reference to measurement procedure is correct.		
8.16	Will the data and parameters result in a conservative estimate of emissions reductions?	/1/ /6/ /9/ /10/	Yes. The source of the data is appropriate, and the data and parameters will result in a conservative estimate of emission reductions.	OK	OK
	Ex-ante calculation of emission reductions	PDD B.6.3			
8.17	Is the projection based on the same procedures as used for future monitoring?	/1/ /6/	Yes. The projection is based on the same procedures as used for future monitoring. This has been verified against the monitoring plan.	OK	OK
8.18	Are the GHG calculations documented in a complete and transparent manner?	/1/ /6/	Yes. The calculations are completely and transparently documented. This has been verified against the data used for the calculation.	OK	OK
8.19	Are detailed calculations provided in a traceable spreadsheet showing relevant information?	/1/ /6/	Yes, a traceable spreadsheet is provided for DOE's review, and has been verified.	OK	OK
8.20	Can the calculation of baseline emissions be replicated using the data and parameters supplied in the PDD?	/1/ /6/	Yes. This has been verified using the data provided.	OK	OK
8.21	Is the data provided in this section consistent with data as presented in other chapters of the PDD?	/1/ /6/	Yes. This has been cross-checked.	OK	OK
	Summary of ex-ante estimation of emission reductions	PDD B.6.4			
8.22	Is the form/ table required for the indication of projected emission reductions correctly applied? And is the data provided in this section consistent with data as presented in other chapters of the PDD?	/1/ /6/	Yes, the table is correctly applied and the numbers have been cross checked against the other sections of the PDD.	OK	OK
8.23	Is the projection in line with the envisioned time schedule for the project's implementation and the indicated crediting period?	/1/	Yes.	OK	OK
9.	Monitoring Plan (CDM VVM EB 55 (para.122-124))	PDD B.7			
	<i>(a) Compliance of the MP with the methodology</i>				
9.1	Are all necessary parameters required for the type of project by the methodology and applicable tools contained in the monitoring plan?	/1/ /6/	The parameter required to be monitored by the methodology is Electricity supplied by the project activity to the grid. The PDD describes the monitoring of two parameters:	OK	OK

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	Checklist Question	Reference	Comment	Draft Conclusion OK/CAR/CL	Final Conclusion OK/ NOT OK
			Parameter (1): power delivered to the grid; and Parameter (2): the electricity consumed by the proposed project, which is imported from the grid. From these two parameters it is possible to calculate net electricity supplied by the project activity to the grid. All parameters are therefore contained in the monitoring plan.		
9.2	For each parameter, is the: a. Title in line with methodology? b. Data unit correctly expressed? c. Parameter appropriately described? d. Source clearly referenced? e. Correct value provided for estimation? f. Has this value been verified? g. Measurement method correctly described? h. Correct reference to standards? i. Indication of accuracy provided? j. QA/QC procedures described? k. QA/QC procedures appropriate?	/1/ /6//31/	EG_{facility,y} a. Yes. The title is inline with the methodology. This has been cross-checked against the methodology, ACM0002. b. Yes. The data unit is MWh/yr, which is correct. c. Yes. The parameter was appropriately described, which is the quantity of net electricity generation supplied by the project plant/unit to the grid in year y. d. Yes. The source was clearly referenced. e. Yes. Ex ante estimated value is 461,464 MWh, which is consistent with the rest of the PDD. f. Yes. The value is consistent with the data presented in other chapters of the PDD and the FSR. g. Yes. The measurement methods is correctly described (calculated as the difference of exported electricity and imported electricity). - h.No. CL 8: The project participant needs to provide the referenced industrial standard in the PDD. CL 9: The project participant would need to address the accuracy for the metering equipment. j. Yes. QA/QC procedure is described. This has been verified against the monitoring methodology. k. Yes. QA/QC in line with the methodology. ERM CVS confirm that the QA/QC procedure is appropriate. EG_{export,y} a. The parameter is included to provide for conservative calculation of EG_{facility,y}. This is considered appropriate. b. Yes. The data unit is MWh, which is correct.	CL8 CL9	OK

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	Checklist Question	Reference	Comment	Draft Conclusion OK/CAR/CL	Final Conclusion OK/ NOT OK
			c. Yes. The parameter was appropriately described, which is the Annual electricity exported to the grid. d. Yes. The source was clearly referenced. e. Yes. Estimated value of annual electricity exported to the grid is 461464 MWh, which is consistent with the rest of the PDD. f. Yes. The value is consistent with the data presented in other chapters of the PDD and FSR. g. Yes. The measurement methods are correctly described. This has been verified against the monitoring plan. h. No. Refer to CL 8 i. Refer to CL 9 j. Yes. QA/QC procedure is described. This has been verified against the monitoring methodology. k. Yes. QA/QC in line with the methodology. ERM CVS confirm that the QA/QC procedure is appropriate. EG_{import,y} a. The parameter is included to provide for conservative calculation of EG_{facility,y}. This is considered appropriate b. Yes. The data unit is MWh, which is correct. c. Yes. The parameter was appropriately described, which is the Annual electricity imported from the grid. d. Yes. The source was clearly referenced. e. Yes. Estimated value of annual electricity exported to the grid is 0 MWh, which is consistent with the rest of the PDD. f. Yes. The value is consistent with the data presented in other chapters of the PDD and FSR. g. Yes. The measurement methods is correctly described. This has been verified against the monitoring plan. h. No. Refer to CL 8 i. Refer to CL 9 j. Yes. QA/QC procedure is described. This has been verified against the monitoring methodology. k. Yes. QA/QC in line with the methodology. ERM CVS confirm that the QA/QC procedure is appropriate.		
9.3	Do all means/ methods of monitoring described in the	/6/	Yes. This has been verified against the monitoring manual and	OK	OK

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	Checklist Question	Reference	Comment	Draft Conclusion OK/CAR/CL	Final Conclusion OK/ NOT OK
	plan comply with the requirements of the methodology?		the methodology, ACM0002.		
	<i>(b) Implementation of the MP</i>				
9.4	Are the arrangements described in the plan feasible and practical within the project design? How has this been verified (review procedures, interviews, project plans, and physical inspection)?	/1/, IV	Yes. The monitoring plan stated in the PDD is feasible and practical considering the type of project activity. This has been verified by reviewing the monitoring plan.	OK	OK
9.5	Is the operational and management structure clearly described and in compliance with the envisioned situation? Are responsibilities and institutional arrangements for data collection and archiving clearly provided?	/1/	Yes. It is clearly described in the PDD, which individual's roles/responsibilities clearly identified.	OK	OK
9.6	Is necessary equipment in place or readily available? Are calibration requirements feasible?	/1/ /6/	Yes, bi-directional main meter and backup meter are installed for the monitoring of net electricity generated to the grid, which are easily available in China. The calibration will be conducted according to the relevant national electric industry standards and regulations by an independent third party. Relevant monitoring equipment is readily available in the host country.	OK	OK
9.7	Does the monitoring plan provide current good monitoring practice?	/1/ /11/	Yes, the project developer has experience in conducting and managing monitoring for the same type of project activities, which can be considered having the knowledge of current good monitoring practice. All data collected will be archived electronically and in paper records and be kept for 2 years after the end of the crediting period, but it should be stated that 'data monitored and required for verification and issuance will be kept for two years after the end of the crediting period or the last issuance of CERs for this project activity, whichever occurs later', according to the EB 'Guidelines for completing the Project Design Document'. All measurements will be conducted with calibrated measurement equipment according to relevant industry standards in China. CAR 3: The statement of 'data monitored and required for verification and issuance will be kept for two years after the end of the crediting period or the last issuance of CERs for this project activity, whichever occurs later' is not included in the PDD.	CAR-3	OK
9.8	If applicable: Does annex 4 provide useful information enabling a better understanding of the envisioned monitoring provisions?		Not applicable	N/A	N/A
9.9	Are the means of implementation of the monitoring plan,	/1/ /11/	Yes. Detailed monitoring and QA/QC procedures stated in the	OK	OK

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	Checklist Question	Reference	Comment	Draft Conclusion OK/CAR/CL	Final Conclusion OK/ NOT OK
	including data management and QA/ QC procedures, sufficient to ensure that the emission reductions achieved can be reported ex-post and verified?		monitoring plan are sufficient to ensure that the emission reductions achieved can be reported ex-post and verified		
9.10	In the DOE's opinion, is the project participant able to implement the monitoring plan?		Yes. The project developer has experience in conducting and managing monitoring for the same type of project activities, such as hydropower and wind power projects.	OK	OK
10.	Sustainable Development (CDM VVM EB 55 (para.125-127))				
10.1	Does the Letter of Approval from the Host Party confirm that the project activity contributes to the sustainable development of that country?	/1/ /11/	No LOA yet available, hence this confirmation is lacking. Please see CAR 1	CAR 1	OK
11.	Environmental Impacts (CDM VVM EB 55 (para.131-133))	PDD D.			
11.1	Has an analysis of the environmental impacts of the project activity been undertaken? How is this evidenced?	/1/ /5/	Yes, this has been verified with the Environmental Impact Assessment report prepared by Environmental Quality Assessment and Research Center of Lanzhon University and the approval of EIA issued by EPB of Gansu Province. No unacceptable adverse environmental impact was identified in the EIA.	OK	OK
11.2	In accordance with the laws and regulations in the Host Country, does this project require an EIA? Has an EIA been conducted for this project? Is this EIA valid for the current project? Has this EIA been approved? How has this been verified?	/1/ /5/	Yes. An EIA for the proposed project was prepared by Environmental Quality Assessment and Research Center of Lanzhon University dated April 2008 and approved by Gansu EPB dated 6 August 2008.	OK	OK
11.3	Does the environmental analysis undertaken and presented for the project activity include an analysis of transboundary impacts? Are any transboundary impacts likely?	/1/ /5/	There are no trans-boundary impacts associated with the project	OK	OK
11.4	Is the analysis in the PDD fully consistent with the findings of the EIF? Are all significant impacts and mitigation measures identified in the EIF mentioned in the PDD?	/1/ /5/	The impacts on air, water, noise and ecological environment have been discussed in the PDD for both the construction period and operation period, and environmental protection measures have been described in PDD. The environmental impact analysis section in the PDD is consistent with the EIA and its approval.	OK	OK
11.5	Does the analysis conclude that the project will create any unacceptable adverse environmental impacts?	/1/ /5/	No unacceptable adverse environmental impacts have been identified according to the EIA and approval.	OK	OK
12.	Local Stakeholder Consultation	PDD E.			

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	Checklist Question	Reference	Comment	Draft Conclusion OK/CAR/CL	Final Conclusion OK/ NOT OK
	(CDM VVM EB 55 (para.128-130))				
12.1	Have comments from relevant stakeholders been invited prior to the publication of the PDD on the UNFCCC website? How has this been verified?	/1//26/	Yes. A stakeholder consultation survey was carried out on 28 April 2009 by means of questionnaires. The representatives include the local government officer and local residents, etc.	OK	OK
12.2	Have all relevant local stakeholders been invited? How has this been assessed?: a) Have appropriate media been used to invite comments by local stakeholders? b) Have all stakeholder groups has access to information? c) Have all stakeholder groups had a reasonable chance to comment?	/1/ /5//26/	Yes. a. Yes. 45 copies of questionnaires were sent to local stakeholders on 28 April 2009 for public consultation. b. Yes. Given that the Project is located in an area with a low population density, and therefore impacts few local residents, the public consultation covered those stakeholder groups, including officials and local residents that may be involved in or affected by the Project. c. Yes. The public consultation was conducted at the initial Project planning stage; thus, all stakeholder groups had a reasonable chance to comment. The DOE interviewed several project stakeholders on-site and confirmed that the consultation took place as described in the PDD.	OK	OK
12.3	Is the summary of comments received as provided in the PDD complete? (what has been done to check this i.e. Document review etc)	/1/ /5//26/	Yes. The PDD includes a summary of the public consultation. The information has been checked against the copies of survey questionnaire provided for review. CL 10 Please clarify the correct numbers of questionnaires delivered and received.	CL 10	OK
12.4	Has due account been taken of any stakeholder comments received and is this adequately and clearly described in the PDD?	/1/ /5//26/	Yes. According to the survey questionnaire, all surveyed stakeholders were reflected in the summary described in the PDD. No negative comments were raised, which was verified by checking the questionnaires.	OK	OK
12.5	In the DOE's opinion, is the local stakeholder consultation process that has been conducted adequate?	/1/ /5//26/	Yes. The public consultation process for the Project is consistent with the common practice for public consultation in China.	OK	OK

Appendix C: REMEDIATION FORM

Corrective Action Requests (CARs), Clarification Requests (CLs), Forward Action Requests (FARs) and Minor Issues

Corrective action requests	Reference to checklist question	Summary of project participants' response	Final conclusion
CAR 1: The LoAs from host party and Annex I party have not been provided yet.	5.2-5.7 10.1	The LoAs have been provided to DOE.	ERM CVS confirm that the LoAs from the host /24/ and Annex I /27/ country are available. CAR 1 is closed.
CAR 2 In order to determine the starting date of project, the project participant would need to provide the construction contract to determine the starting date	7.1 7.3	The starting date of the project has been corrected to 24 October 2009, since the Engineering and Construction Contract was signed on 24 October 2009. The relevant documents have been provided.	The validation team cross checked the construction contract and confirmed the signed date, which is 24 October 2009 /20/, the earliest starting date of the proposed project among the equipment purchase contracts /19/ and main construction contracts /20/. CAR 2 is closed.
CAR 3: The statement of 'data monitored and required for verification and issuance will be kept for two years after the end of the crediting period or the last issuance of CERs for this project activity, whichever occurs later' is not included in the PDD.	9.7	The statement has been corrected in part B.7.2 of the PDD.	The statement was revised as "All the electronic and paper documents will be archived during the crediting period and two years after the end of the crediting period or the last issuance of CERs, whichever occurs later." CAR 3 is closed.

Clarification requests	Reference to checklist question	Summary of project participants' response	Final conclusion
CL 1: The following technical information is not provided in section A.4.3 of the PDD, as required by the PDD guidelines: (1) the lifetime of the project equipment and the reference for the project operational lifetime should be added in section C.1.2; (2) monitoring equipments; (3) technology transfer;	4.5-4.6 7.17 7.18 7.19	(1) The life time has been added into the Section C.1.2 of the PDD. (2) Monitoring Equipments have been added. (3) Technology transfer has been added. (4) The load factor has been added. (5) GHGs emission sources have been added. (6) Effective electricity coefficient has been added.	ERM CVS check the FSR /4/, ERM CVS confirm that the following technical information have been updated in the Section A.4.3 of the PDD which are in line with the FSR /4/. (1) the lifetime of the project is 22 years (2) monitoring equipments is electricity meter(s);

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Clarification requests	Reference to checklist question	Summary of project participants' response	Final conclusion
(4) load factor; and (5) GHGs emission sources. (6) effective electricity coefficient			(3) No technology transfer is involved; (4) load factor is 26.2% (5) GHGs emission sources is CO ₂ . (6) The annual power generation of the project is 461,464 MWh, the annual net power supplied to the grid is 461,464 MWh and the electricity consumption is assumed as 0 MWh. CL 1 is closed.
CL 2: Please provide training material for project operation.	4.10	The operation staff training plan have been provided. The training date is 20/10/2010	The training material /40/ has been provided to ERM CVS. It is considered that the training material is appropriate to ensure monitoring of emission reductions ex-post. CL 2 is closed.
CL 3: Please include the flows of mass/energy and monitoring variables in the diagram in section B.3 of the PDD, as required by the PDD Guidelines.	6.10	The diagram in section B.3 has been corrected accordingly.	The flows of mass/energy and monitoring variables were properly added in the diagram in PDD B.3. CL 3 is closed.
CL 4: The validation team has verified the investment cost against the approved FSR and it is deemed to be the same, however, please provide further evidence to cross check the investment cost.	7.19 7.22 7.20	The total investment cost is 1908.03 million RMB Yuan. The turbine purchase contracts and the EPC contract have been provided, in which the total fee of the three contracts is 1840.05647 million RMB Yuan which is only 3.56% lower than the investments cost. Hence it is conservative. See page 212 and page 213 of the FSR report.	In order to further cross check and confirm the investment costs, the principal investment related third party contracts were reviewed /19//20/, the total contracted value is approximately 1,840 million RMB. ERM CVS was able to review the cost breakdown of the same components in the FSR and the overall variance between the contracts and the FSR estimation of the cost is 3.56%, which shows that the estimation of the FSR is reasonable given that the project was still under construction and further expenditure was still yet to be incurred. Therefore the estimated value in the FSR is considered reasonable. CL 4 is closed.

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Clarification requests	Reference to checklist question	Summary of project participants' response	Final conclusion
CL 5: The inclusion of 100% of maintenance costs for the last two years shall be explained, bearing in mind that electricity generation declines in these years.	7.30	The electricity generation is expected to be lower than the last two years of operation, i.e. after 20 years of operation, because of the wear and tear on the equipment. In order to generate certain electricity in the last two years, the maintenance would be kept the same as the previous years. It based on the experience of Northwest Hydro Consulting Engineers who prepared the FSR. Furthermore, the OM cost in the IRR sheet was sourced from the approved FSR and the OM cost during the last two years has little impact to the IRR.	The inclusion of 100% of maintenance costs for the last two years is in line with the statement of the FSR /4/. Based on ERM CVS's local and sectoral knowledge it is considered that after 20 years of operation the generation of the wind turbines may be expected to fall slightly, and the difference is estimated by the independent third party design institute based on their experience and expertise. Full O&M costs are still included because the wind turbines will still be operating and will need maintenance and repair as normal. This situation will impact on the IRR only very slightly. Even if we assume that the maintenance costs for the last two years are zero, the IRR would be 6% which is lower than the benchmark of 8%. So it is considered to be reasonable that the maintenance costs will remain at the same level as previous years. CL 5 is closed.
CL 6: Explanation of the construction loan interests shall be provided.	7.32	According to Section 2.10 of the Economic Evaluation Methods and Parameters of Construction Projects, interest during construction should be assessed in the investment analysis. Interest during construction should be calculated according to the project construction schedule and the investment plan. The construction loan interest of 63.4 million RMB is sourced from the FSR which is approved by the NDRC. The construction loan interests are designed according to the construction schedule and the investment of the proposed project. Supposing the construction loan interests cut to zero, the IRR without CDM would be 5.90% which is 0.09% higher than 5.81%. Hence there is little impact on the IRR.	ERM CVS has checked Section 2.10 of the Economic Evaluation Methods and Parameters of Construction Projects /42/. In this document it is stated that the construction loan interest should be included in the investment analysis. Also, ERM CVS can confirm that the construction loan interest is taken from the approved FSR /4/. This parameter will have a minimal effect on the IRR. Even if it is assumed that the construction loan interest is zero, the IRR will be 5.9% which is lower than the benchmark of 8% /3/. Therefore ERM CVS can confirm that taking into account the construction loan interest during the IRR assessment is

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Clarification requests	Reference to checklist question	Summary of project participants' response	Final conclusion
			reasonable and appropriate in line with host country accounting regulations and also does not have a detrimental impact on the assessment of additionality. CL 6 is closed
CL 7: Please provide the results of the sensitivity analysis on what variations are needed for static investment, O&M costs, tariff and annual output, in order to reach the benchmark for project IRR in the PDD.	7.34	The relevant sensitivity analysis has been added into the PDD.	Sensitivity analysis was correctly added in the PDD section B.5 sub-step 2d, including the results of the sensitivity analysis on what variations are needed for static investment, O&M costs, tariff and annual output to change, in order to reach the benchmark for project IRR in the PDD, respective explanations were also put on. The sensitivity analysis is validated in detail in section 3.5 of the validation report. CL 7 is closed.
CL 8: The project participant needs to provide the referenced industrial standard in the PDD.	9.2	The relevant industrial standard has been provided: ● http://www.newenergy.org.cn/html/0085/5290817781.html ● China Electric Power Yearbook 2008 ● National VAT Law issued by State Administration of Taxation (State Council [2008]538) and VAT policy on Comprehensive Utilization of Resource (http://www.jstax.gov.cn/Page1/StatuteDetail.aspx?StatuteID=8862) and Other Products released by Ministry of Finance and State Administration of Taxation (Cai Shui[2008]156) (http://www.jstax.gov.cn/Page1/StatuteDetail.aspx?StatuteID=8931)	ERM CVS has already checked relevant industrial standards below: ● The development of wind turbine in China (http://www.newenergy.org.cn/html/0085/5290817781.html) /44/ ● China Electric Power Yearbook 2008 /15/ ● National VAT Law issued by State Administration of Taxation /28/ The monitoring plan is in line with the standards. CL 8 is closed.
CL 9: The project participant needs to address the accuracy for the metering equipment.	9.2	The accuracy of the metering equipment has been added into the PDD.	The accuracy of metering equipment was added in PDD section 7.2. The main meter M ₁₁ and the back-up meter M ₁₂ are in accuracy of 0.2s, the back-up line meter M ₂₁ is in accuracy of 0.5. CL 9 is closed.

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Clarification requests	Reference to checklist question	Summary of project participants' response	Final conclusion
CL 10: Please clarify the correct numbers of questionnaires delivered and received.	12.3	The number of questionnaires delivered should be 50 and 50 of them were collected.	The PDD section E.2 was revised, 50 questionnaires were delivered and 50 of them were collected. CL 10 is closed.

Minor Issue	Reference to checklist question	Summary of project participants' response	Final conclusion
Minor Issue 1: Spreadsheet refers to depreciation as "devalue". The correct term shall be used.	7.30	The description 'devalue' has been corrected to 'depreciation'.	The spreadsheet was correctly revised. Minor Issue 1 is closed.

Forward Action Requests	Reference to checklist question	Summary of project participants' response	Final conclusion
No FARs raised.			